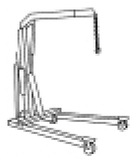
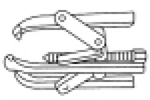
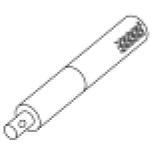
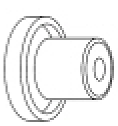
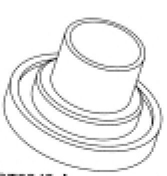
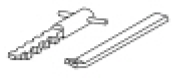
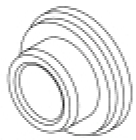
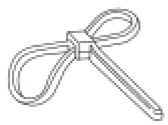
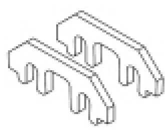
Engine

Ford Ecat Electronic Parts Catalog

Special Tool(s)

 ST1341-A	2,200# Floor Crane, Fold Away 300-OTC1819E
 ST1184-A	3 Jaw Puller 303-D121
 ST3101-A	Bar, Engine Spreader 303-1454
 ST1326-A	Handle 205-153 (T80T-4000-W)
 ST3234-A	Hold Down, Secondary Chain 303-1530
 ST2212-A	Installer, Differential Bearing Cone 205-142 (T80T-4000-J)
 ST2240-A	Installer, Spindle Bearing 205-150 (T80T-4000-S)

 ST1385-A	Remover, Oil Seal 303-409 (T92C-6700CH)
 ST2982-A	Remover, VCT Spark Plug Tube Seal 303-1247/1
 ST1436-A	Strap Wrench 303-D055 (D85L-6000-A)
 ST2979-A	Tool, Camshaft Holding 303-1248

Material

Item	Specification
Motorcraft® Metal Surface Prep ZC-31-B	—
Motorcraft® Silicone Gasket Remover ZC-30-A	—

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

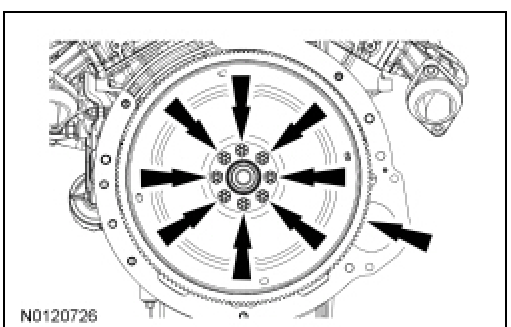
NOTICE: The engine is equipped with a cast aluminum (early build) or stamped steel (late build) crankshaft rear seal retainer plate. All crankshaft rear seal retainer plates must be discarded and a new stamped steel rear seal retainer plate installed.

NOTE: *If the cylinder head(s) is replaced, a new secondary timing chain tensioner will need to be installed.*

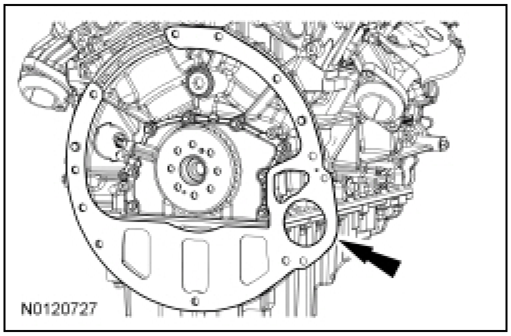
NOTE: *For additional information, refer to the exploded view under the Assembly procedure in this section.*

All engines

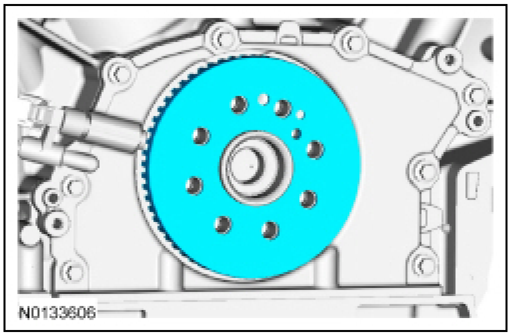
1. Remove the 8 bolts and the flexplate.



2. Remove the spacer plate.

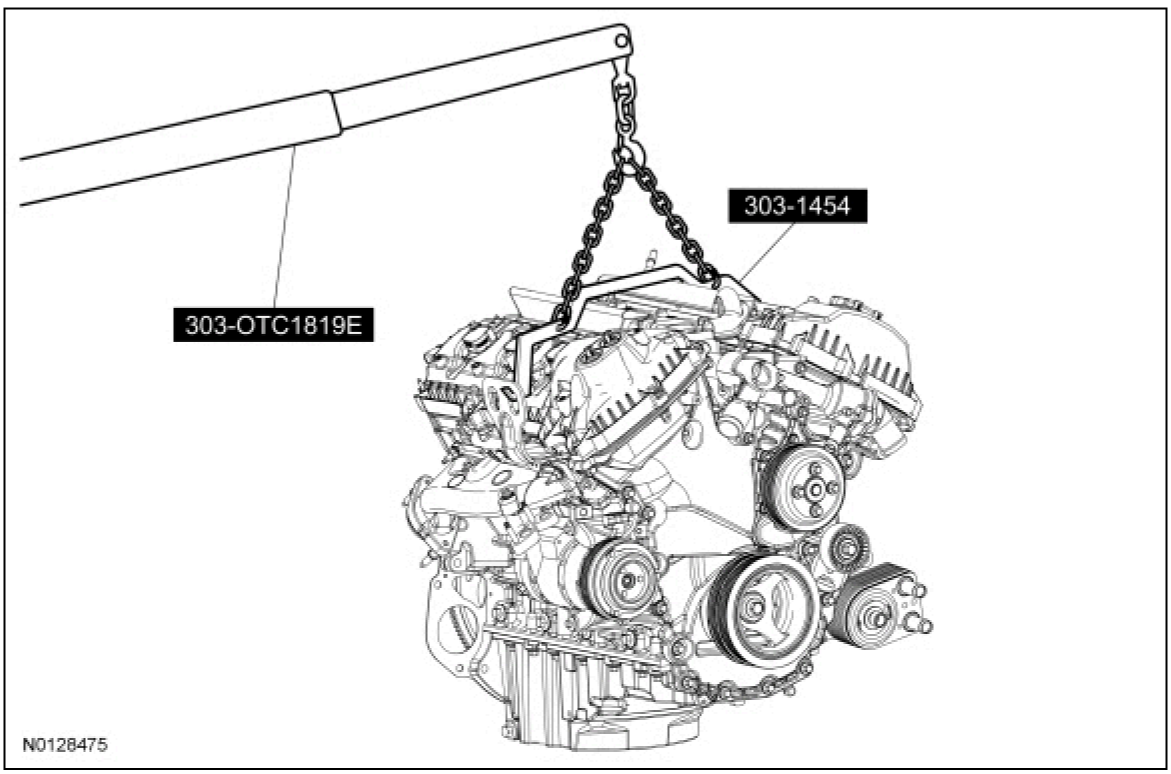


3. Remove the ignition pulse ring.



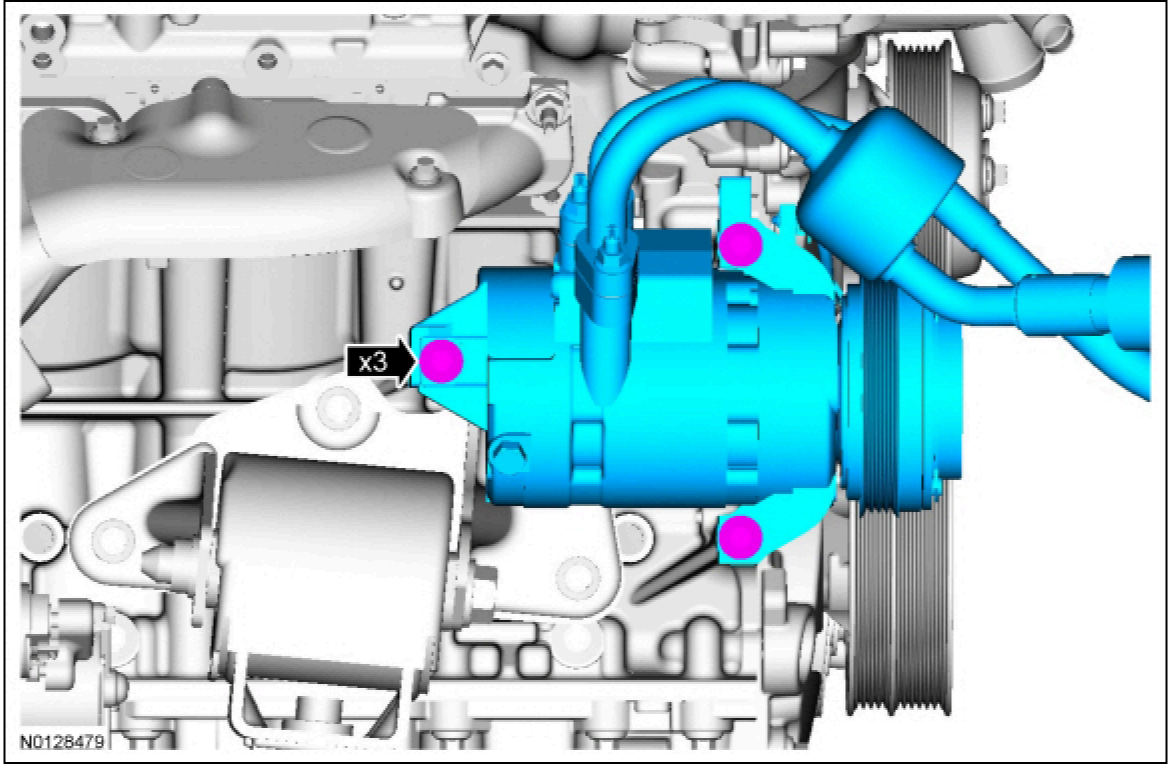
4. **NOTE:** *Install the engine stand bolts into the cylinder block only. Do not install the bolts into the oil pan.*

   Using the Heavy Duty Floor Crane and Spreader Bar, mount the engine on a suitable engine stand.

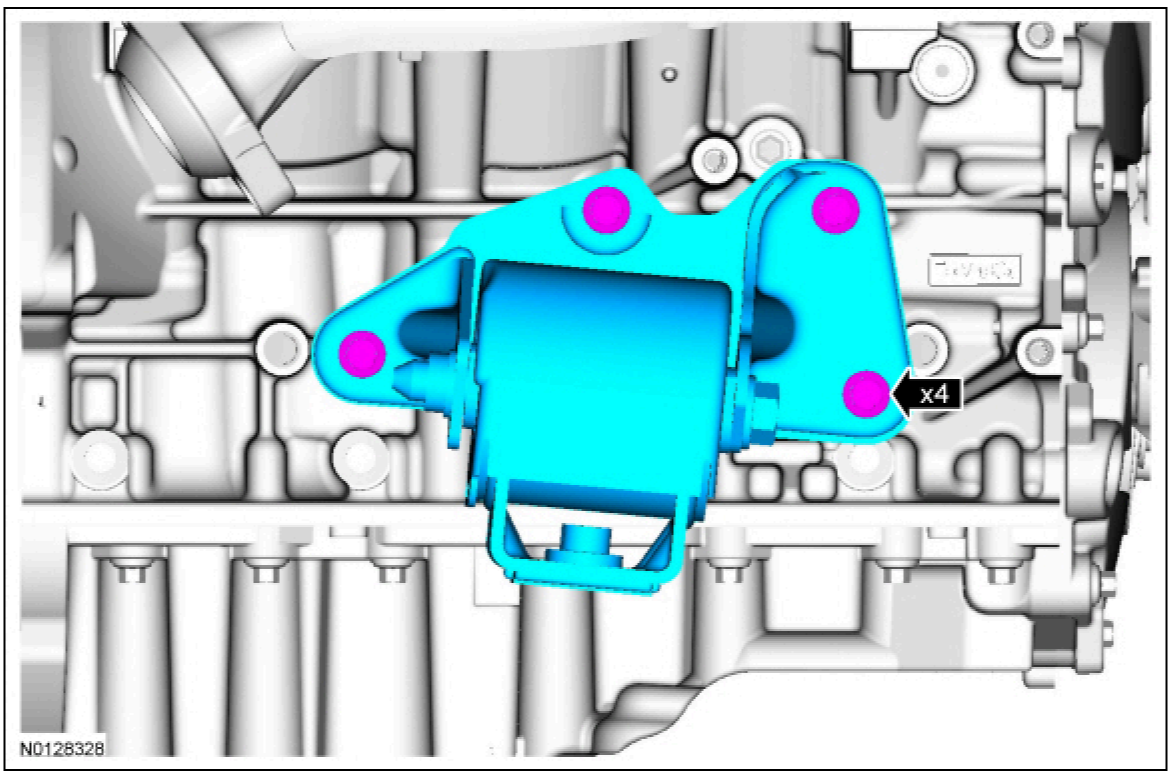


5. Remove the LH and RH engine lift eyes.

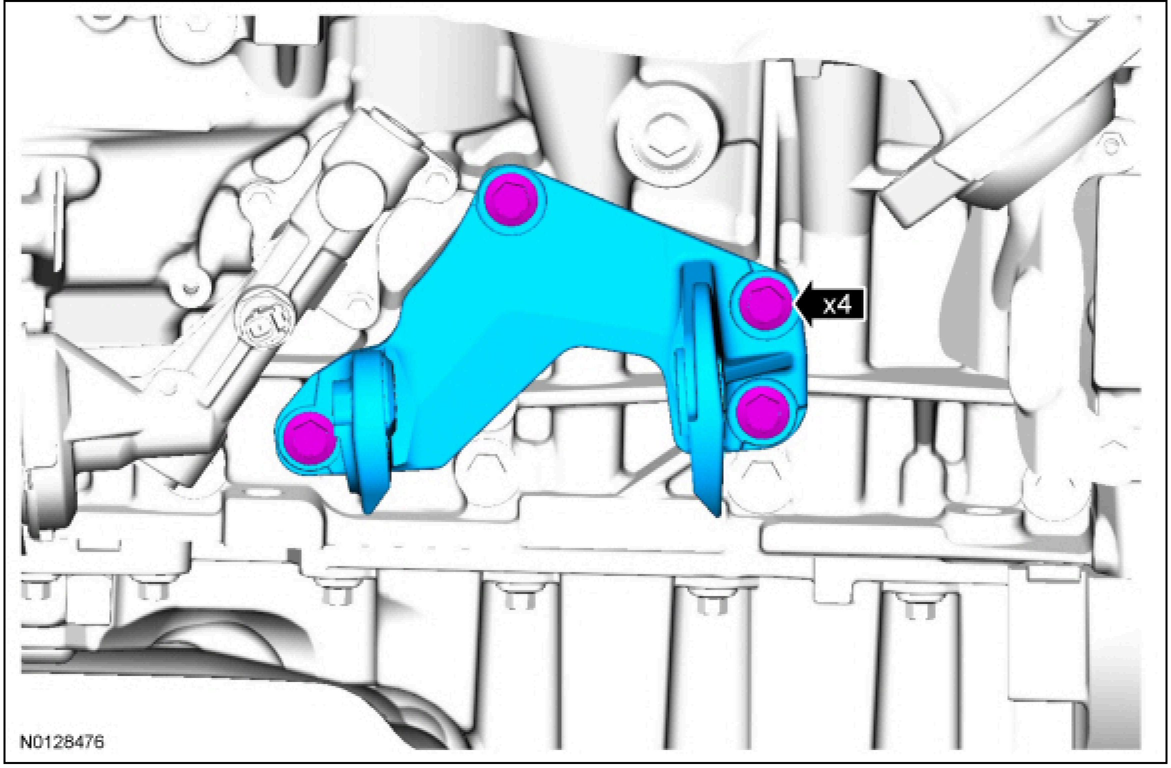
6. Remove the 3 bolts and the A/C compressor .



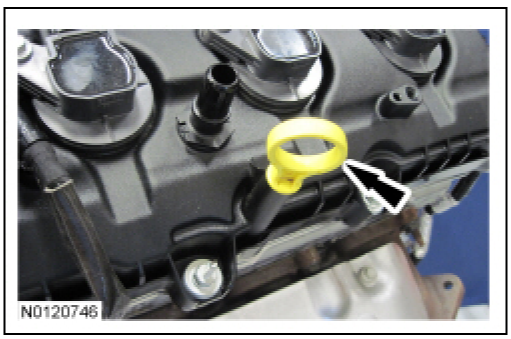
7. Remove the 4 bolts and the RH engine support insulator.



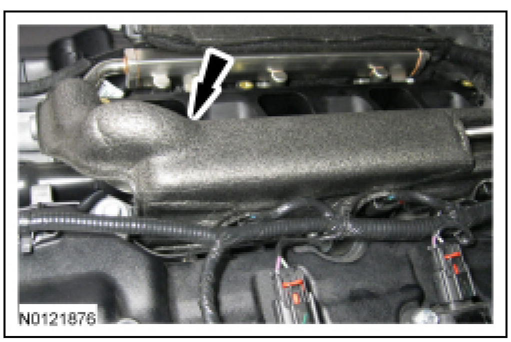
8. Remove the 4 bolts and the LH engine support insulator-to-cylinder block bracket.



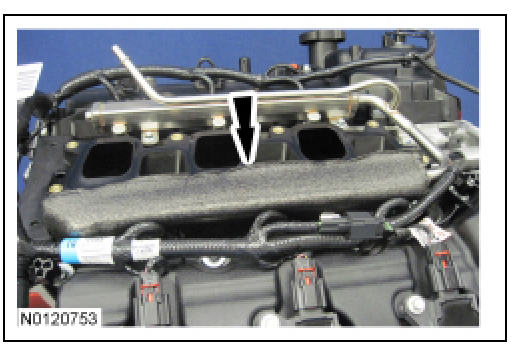
9. Remove the engine oil level indicator.



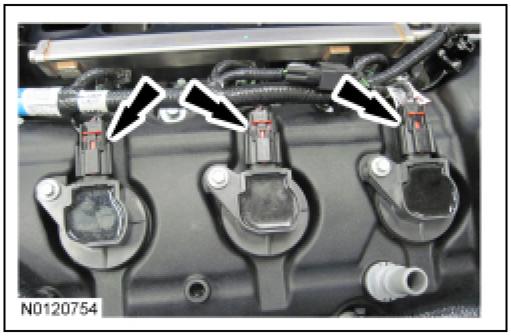
10. Remove the LH fuel rail insulator.



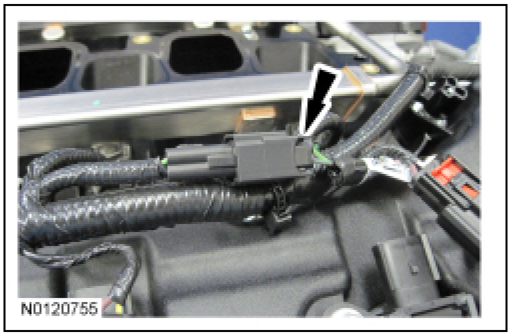
11. Remove the RH fuel rail insulator.



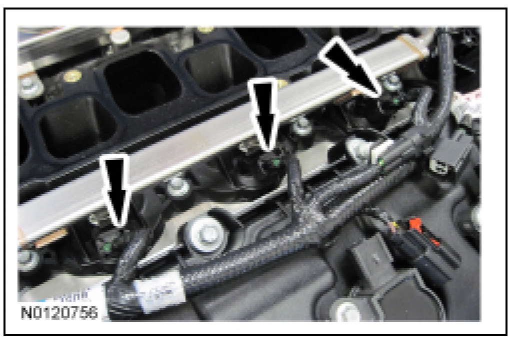
12. Disconnect the 3 RH ignition coil-on-plug electrical connectors.



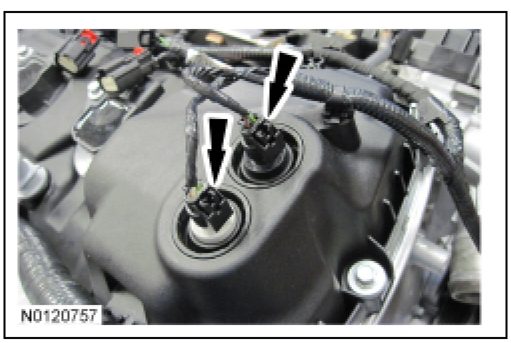
13. Disconnect the Cylinder Head Temperature (CHT) sensor wiring harness jumper electrical connector.



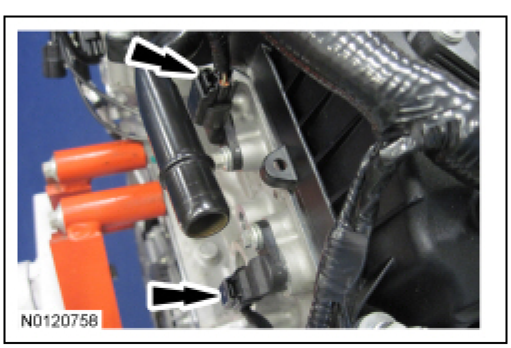
14. Disconnect the 3 RH fuel rail electrical connectors.



15. Disconnect the 2 RH Variable Camshaft Timing (VCT) solenoid electrical connectors.



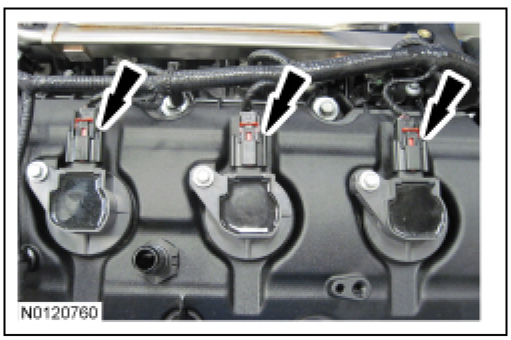
16. Disconnect the 2 RH Camshaft Position (CMP) sensors electrical connectors.



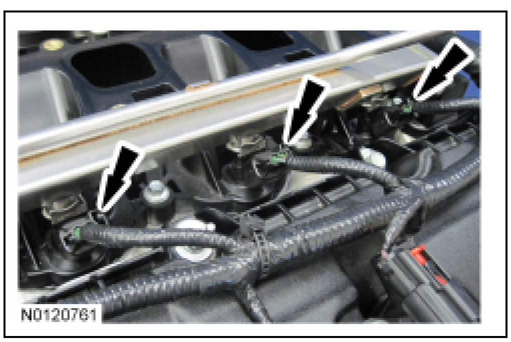
17. Detach all of the wiring harness retainers from the RH cylinder head, valve cover and stud bolts.

18. Disconnect the Engine Oil Pressure (EOP) switch electrical connector.
    - Detach the EOP switch wiring harness retainer from the cylinder head.

19. Disconnect the 3 LH ignition coil-on-plug electrical connectors.

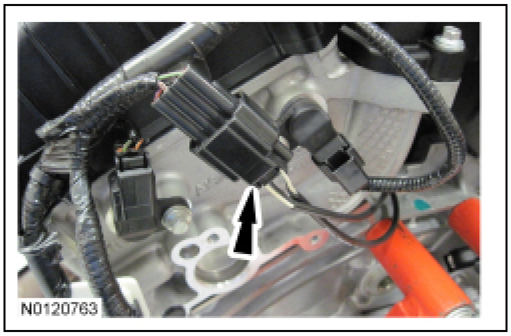


20. Disconnect the 3 LH fuel rail electrical connectors.

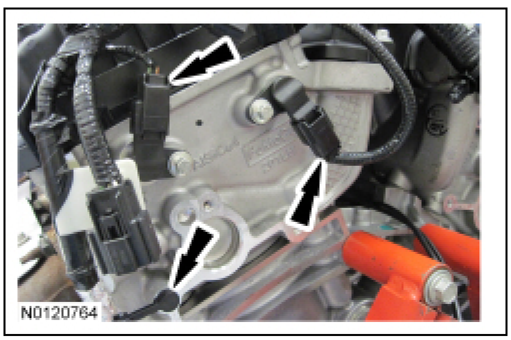


21. Disconnect the 2 LH VCT solenoid electrical connectors.

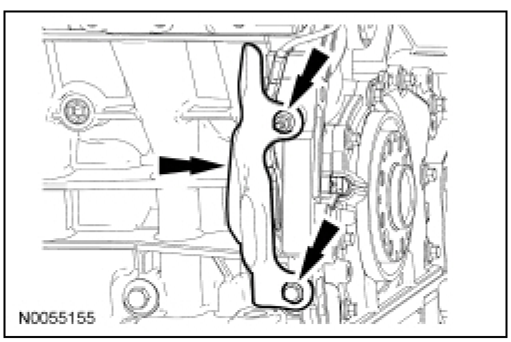
22. Disconnect the Knock Sensor (KS) electrical connector.



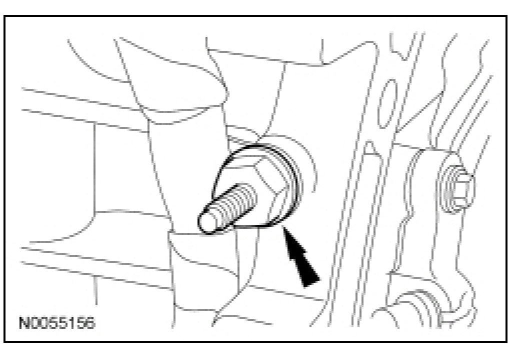
23. Disconnect the 2 LH CMP sensors electrical connectors.
    - Detach the wiring harness retainer from the rear of the cylinder head.



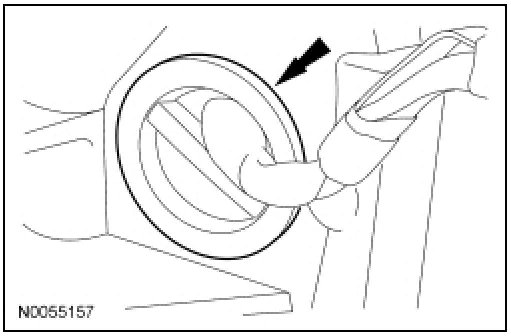
24. Remove the nut, the bolt and the heat shield.



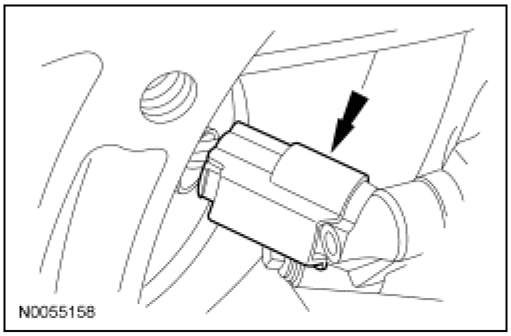
25. Remove the wiring harness retainer stud bolt.



26. Remove the wiring harness grommet.

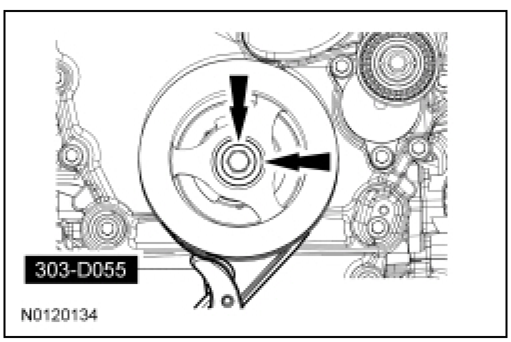


27. Disconnect the Crankshaft Position (CKP) sensor electrical connector.

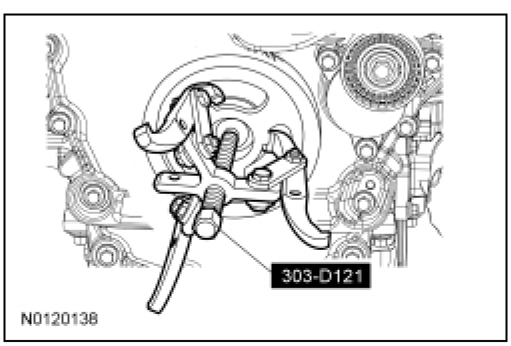


28. Detach all of the wiring harness retainers from the LH cylinder head, valve cover and stud bolts.

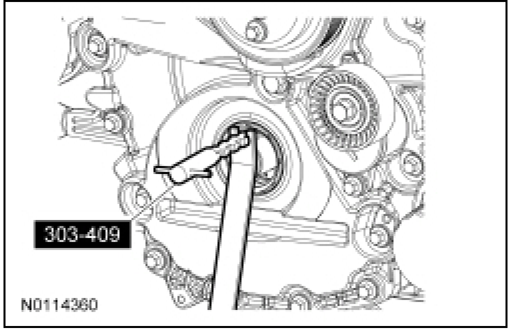
29. Using the Strap Wrench, remove the crankshaft pulley bolt and washer.
    - Inspect the crankshaft pulley sealing surface for nicks or scratches that may effect the crankshaft front seal lip, replace the crankshaft pulley if necessary.
    - Discard the bolt.



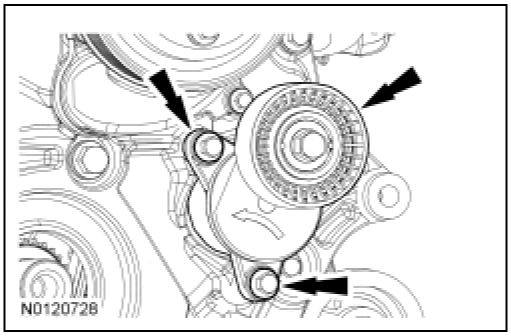
30. Using the 3 Jaw Puller, remove the crankshaft pulley.



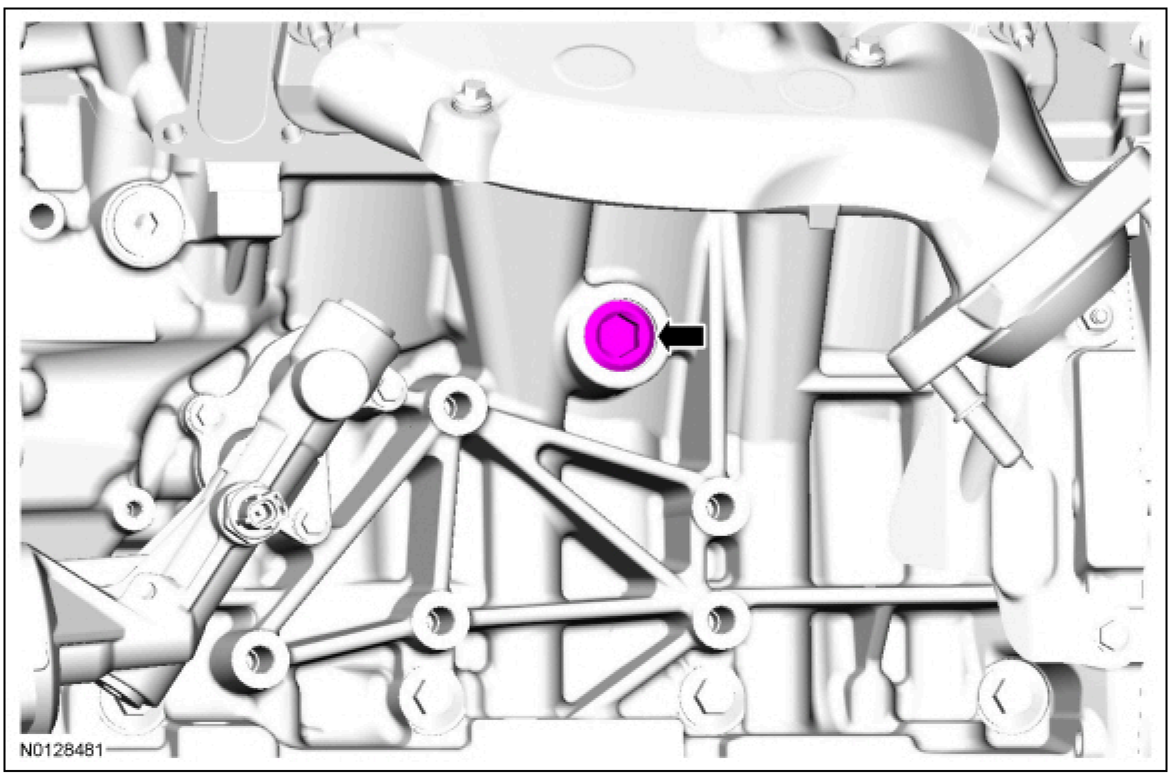
31. Using the Oil Seal Remover, remove and discard the crankshaft front seal.



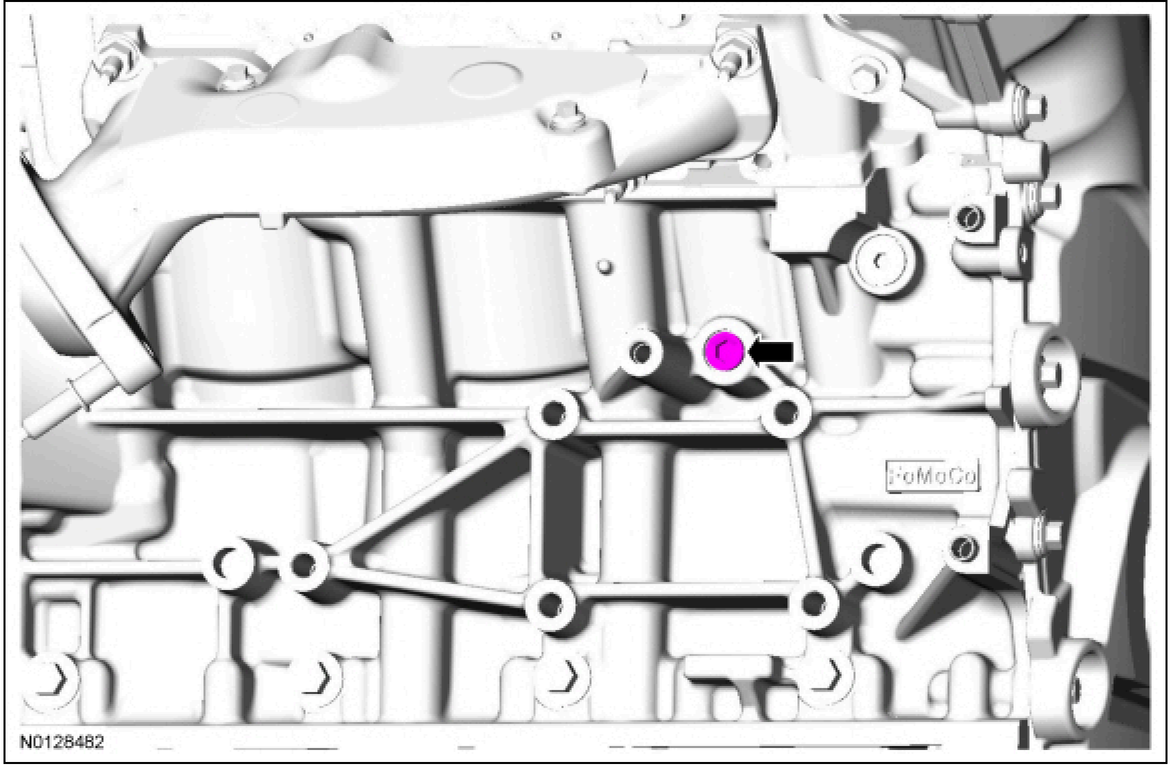
32. Remove the 2 bolts and the accessory drive belt tensioner.



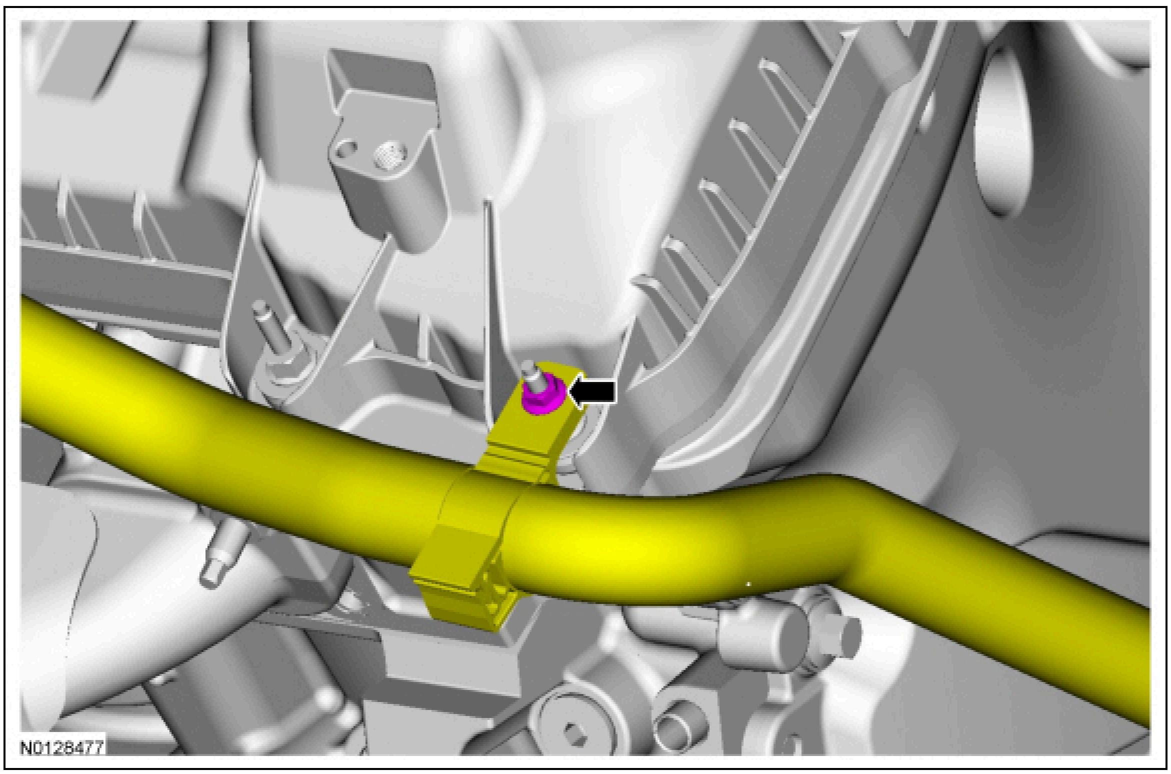
33. Remove the LH cylinder block drain plug.
    - Allow coolant to drain from the cylinder block.



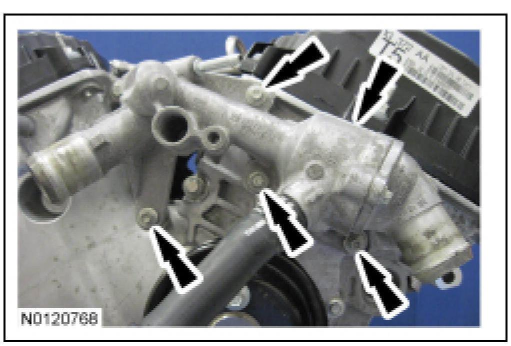
34. Remove the RH cylinder block drain plug.
    - Allow coolant to drain from the cylinder block.



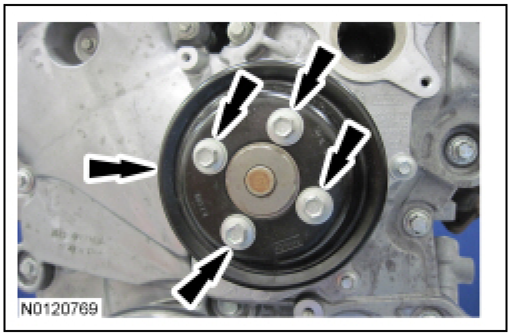
35. Remove the nut and the heater hose bracket from the valve cover stud bolt.



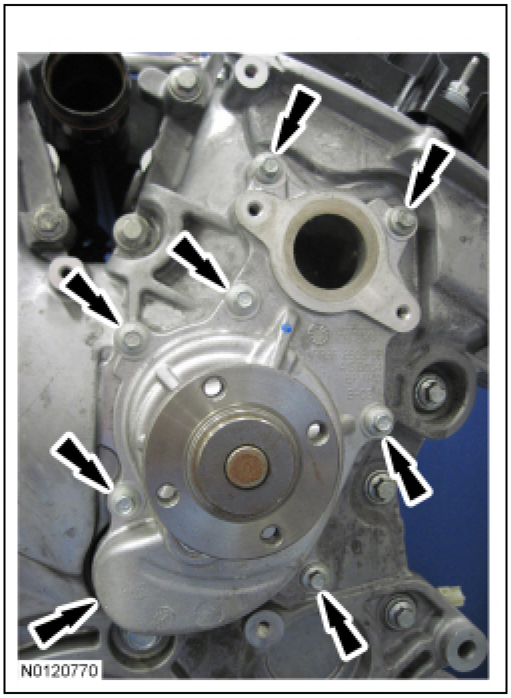
36. Remove the 4 bolts and the thermostat housing.
    - Discard the thermostat housing and coolant inlet tube O-ring seals.



37. Remove the 4 bolts and the coolant pump pulley.

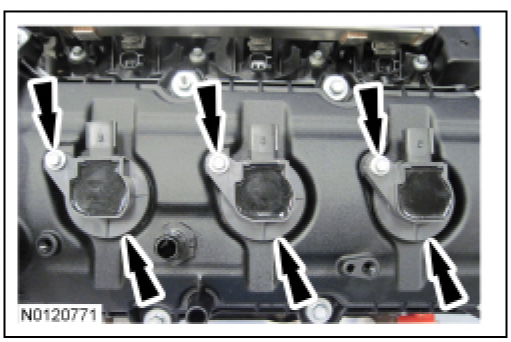


38. Remove the 7 bolts and the coolant pump.
    - Discard the coolant pump gasket and coolant pump O-ring seal.



39. **NOTE:** *When removing the ignition coil-on-plugs, a slight twisting motion will break the seal and ease removal.*

    Remove the 3 bolts and the 3 LH ignition coil-on-plugs.



40. **NOTICE: Only use hand tools when removing the spark plugs, or damage may occur to the cylinder head or spark plug.**

    NOTE: *Use compressed air to remove any foreign material in the spark plug well before removing the spark plugs.*

    Remove the 3 LH spark plugs.

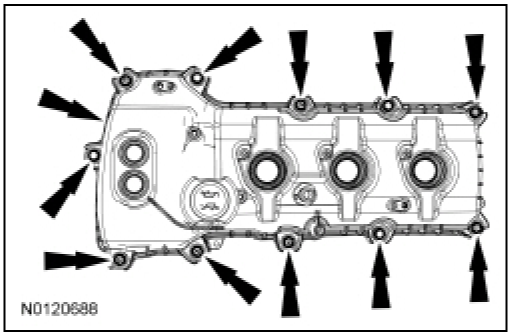
41. **NOTICE: While removing the valve cover do not apply excessive force to the Variable Camshaft Timing (VCT) oil control solenoid or damage may occur.**

NOTICE: If the Variable Camshaft Timing (VCT) oil control solenoid sticks to the VCT seal, carefully wiggle the valve cover until the bond breaks free or damage to the VCT seal and VCT oil control solenoid may occur.

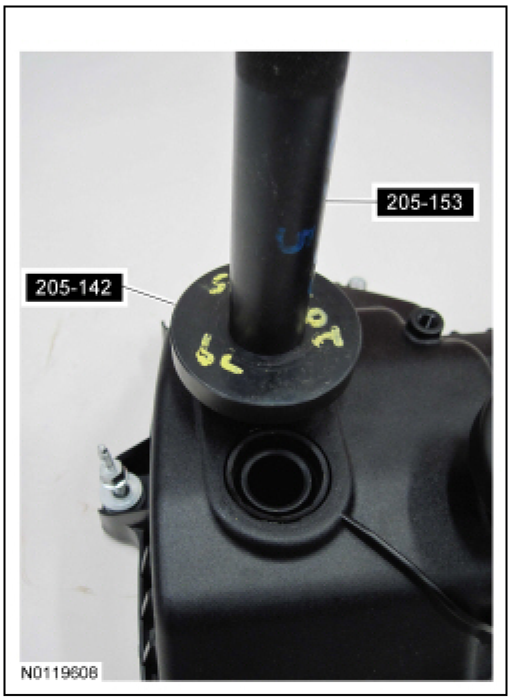
NOTE: *The plastic electrical connector on the VCT oil control solenoid will rotate approximately 12 degrees inside the steel housing, which is normal.*

Loosen the 7 stud bolts, 4 bolts and remove the LH valve cover.

- Discard the gasket.



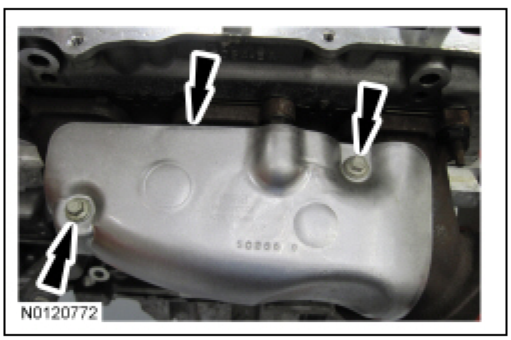
42. Inspect the 2 VCT solenoid seals. Remove any damaged seals.
    - Using the Differential Bearing Cone Installer and Handle, remove the seal(s).



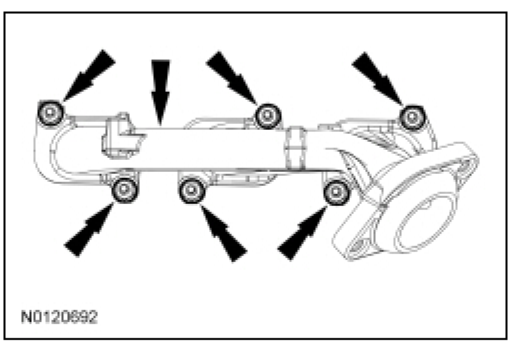
43. Inspect the spark plug tube seals. Remove any damaged seals.
    - Using the VCT Spark Plug Tube Seal Remover and Handle, remove the seal(s).

44. Clean the valve cover, cylinder head and engine front cover sealing surfaces with metal surface prep.

45. Remove the 2 bolts and the LH exhaust manifold heat shield.

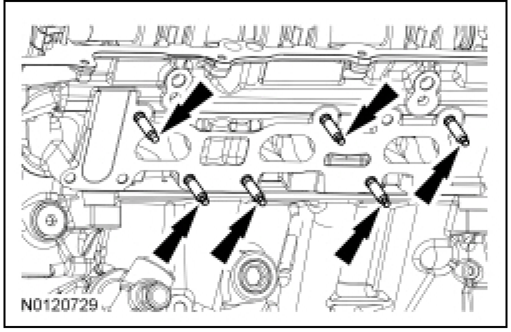


46. Remove the 6 nuts and the LH exhaust manifold.
    - Discard the nuts and exhaust manifold gaskets.



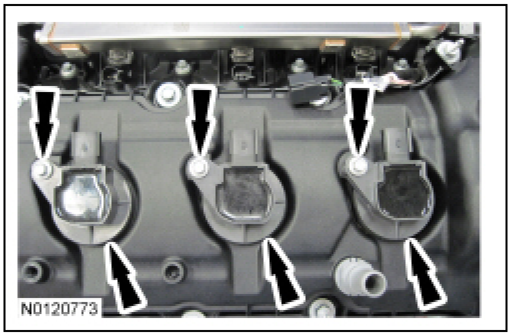
47. Clean and inspect the LH exhaust manifold. For additional information, refer to Section 303-00.

48. Remove and discard the 6 LH exhaust manifold studs.



49. **NOTE:** *When removing the ignition coil-on-plugs, a slight twisting motion will break the seal and ease removal.*

    Remove the 3 bolts and the 3 RH ignition coil-on-plugs.



50. **NOTICE: Only use hand tools when removing the spark plugs, or damage may occur to the cylinder head or spark plug.**

    NOTE: *Use compressed air to remove any foreign material in the spark plug well before removing the spark plugs.*

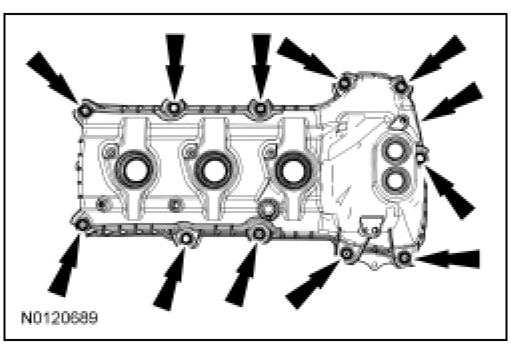
    Remove the 3 RH spark plugs.

51. **NOTICE: While removing the valve cover do not apply excessive force to the Variable Camshaft Timing (VCT) oil control solenoid or damage may occur.**

    NOTICE: If the Variable Camshaft Timing (VCT) oil control solenoid sticks to the VCT seal, carefully wiggle the valve cover until the bond breaks free or damage to the VCT seal and VCT oil control solenoid may occur.

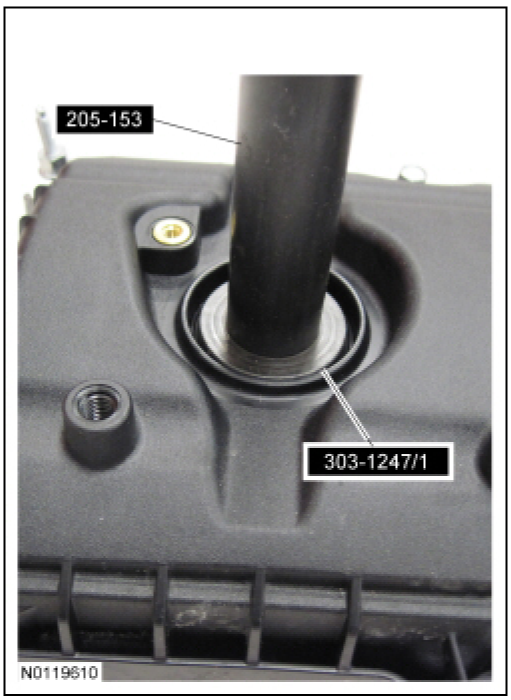
    NOTE: *The plastic electrical connector on the VCT oil control solenoid will rotate approximately 12 degrees inside the steel housing, which is normal.*

    Loosen the 8 stud bolts, 3 bolts and remove the RH valve cover.
    - Discard the gasket.



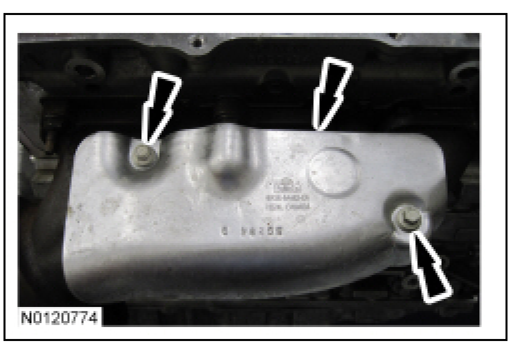
52. Inspect the 2 VCT solenoid seals. Remove any damaged seals.
    - Using the Differential Bearing Cone Installer and Handle, remove the seal(s).

53. Inspect the spark plug tube seals. Remove any damaged seals.
    - Using the VCT Spark Plug Tube Seal Remover and Handle, remove the seal(s).



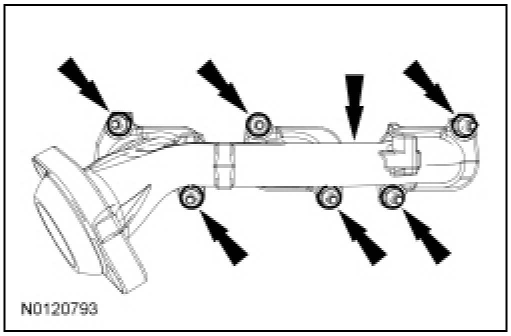
54. Clean the valve cover, cylinder head and engine front cover sealing surfaces with metal surface prep.

55. Remove the 2 bolts and the RH exhaust manifold heat shield.



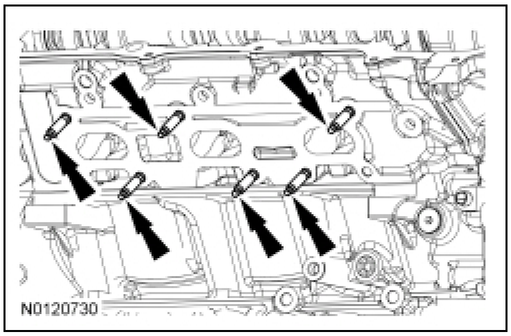
56. Remove the 6 nuts and the RH exhaust manifold.

- Discard the nuts and exhaust manifold gaskets.

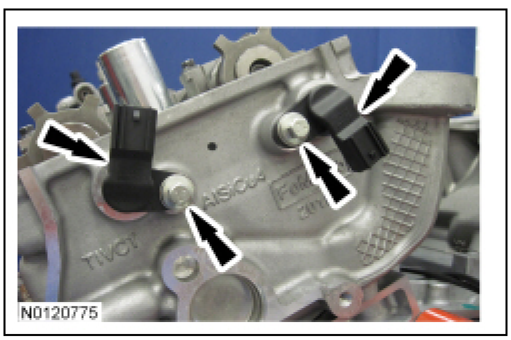


57. Clean and inspect the RH exhaust manifold. For additional information, refer to Section 303-00.

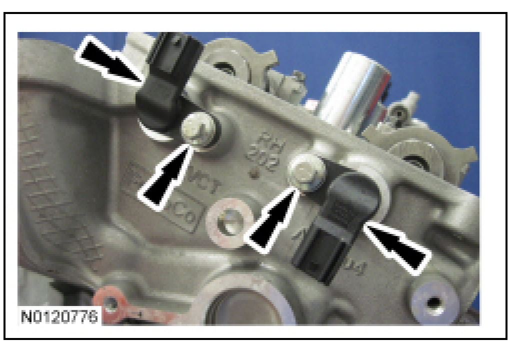
58. Remove and discard the 6 RH exhaust manifold studs.



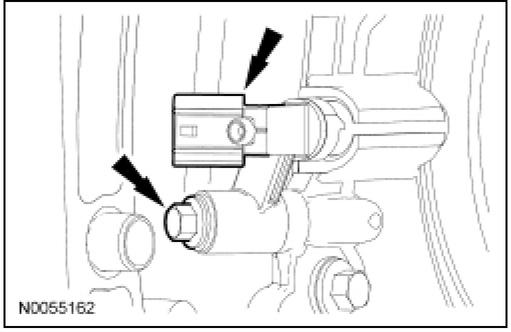
59. Remove the 2 bolts and the LH CMP sensors.



60. Remove the 2 bolts and the RH CMP sensors.

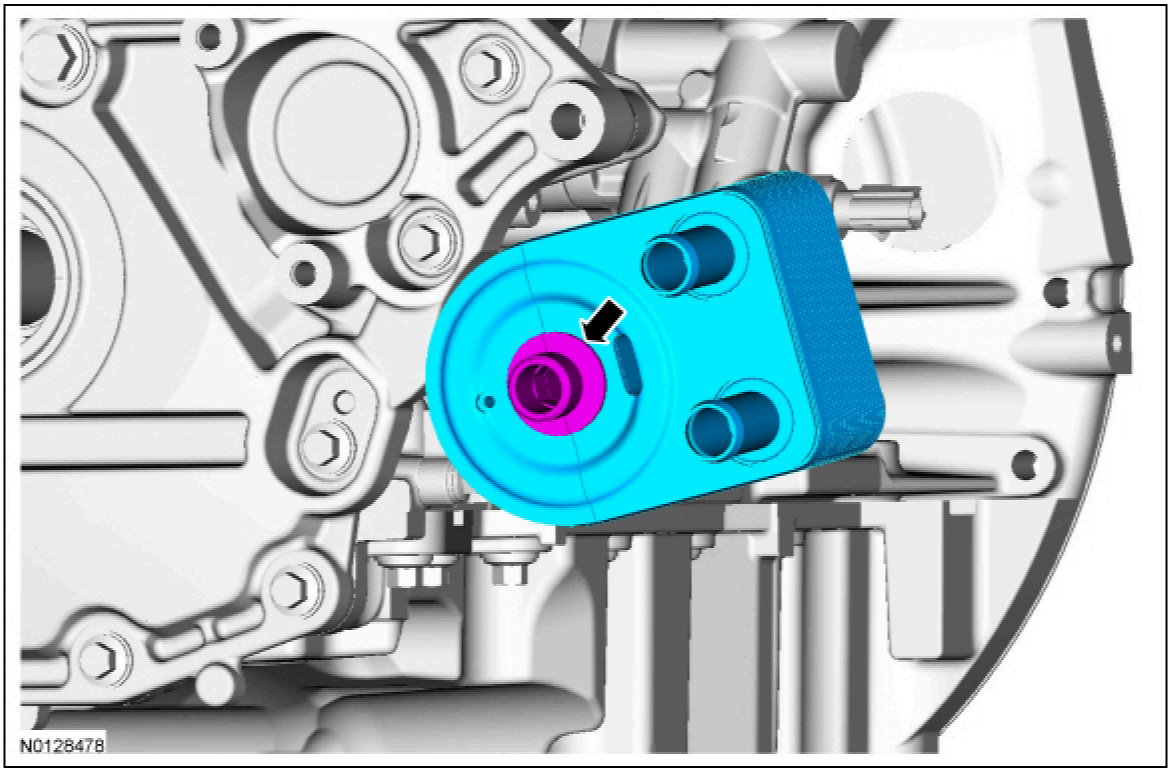


61. Remove the bolt and the CKP sensor.

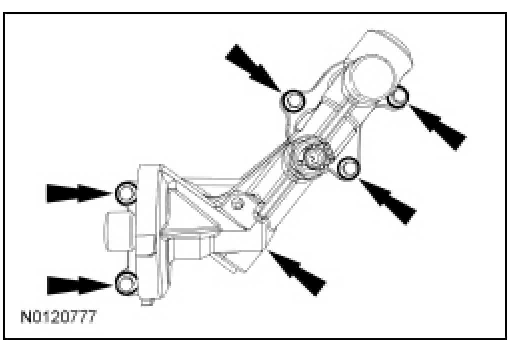


62. **NOTICE: A new oil cooler must be installed or severe damage to the engine can occur.**

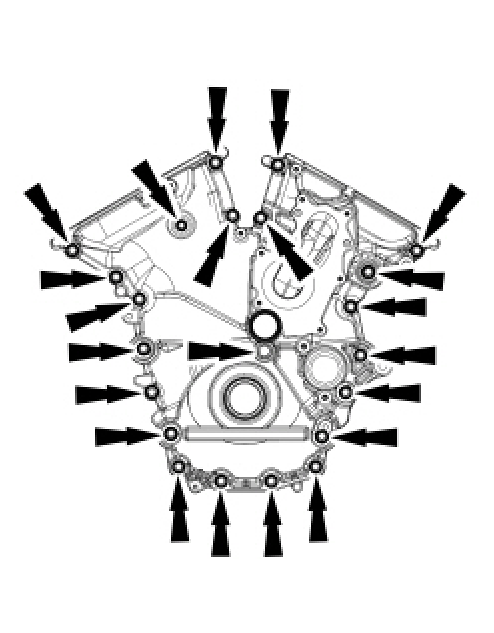
    If equipped, remove the oil cooler mounting bolt and the oil cooler.



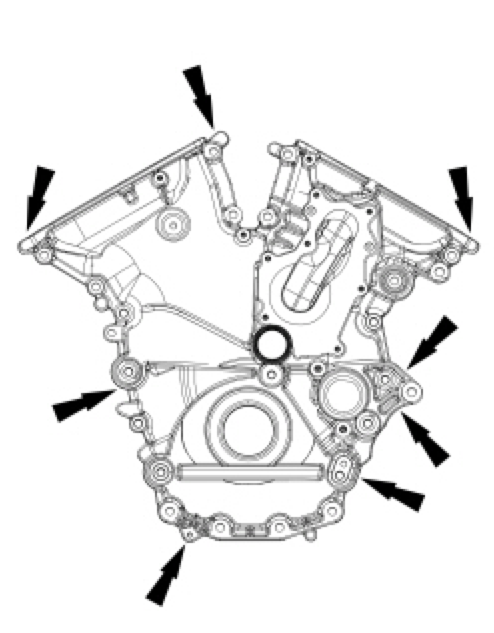
63. Remove the 5 bolts and the oil filter adapter.
    - Discard the gasket.



64. Remove the 22 engine front cover bolts.


N0120120

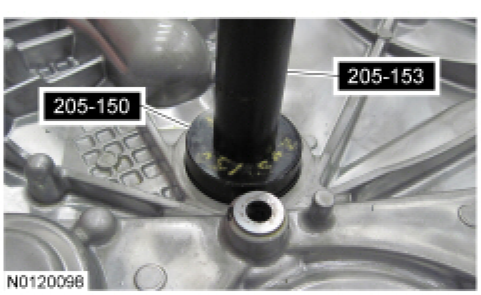
65. Using a pry tool, locate the 7 pry pads shown and pry the engine front cover loose and remove.


N0120690

66. **NOTE:** *Inspect front cover radial seal and replace if necessary.*

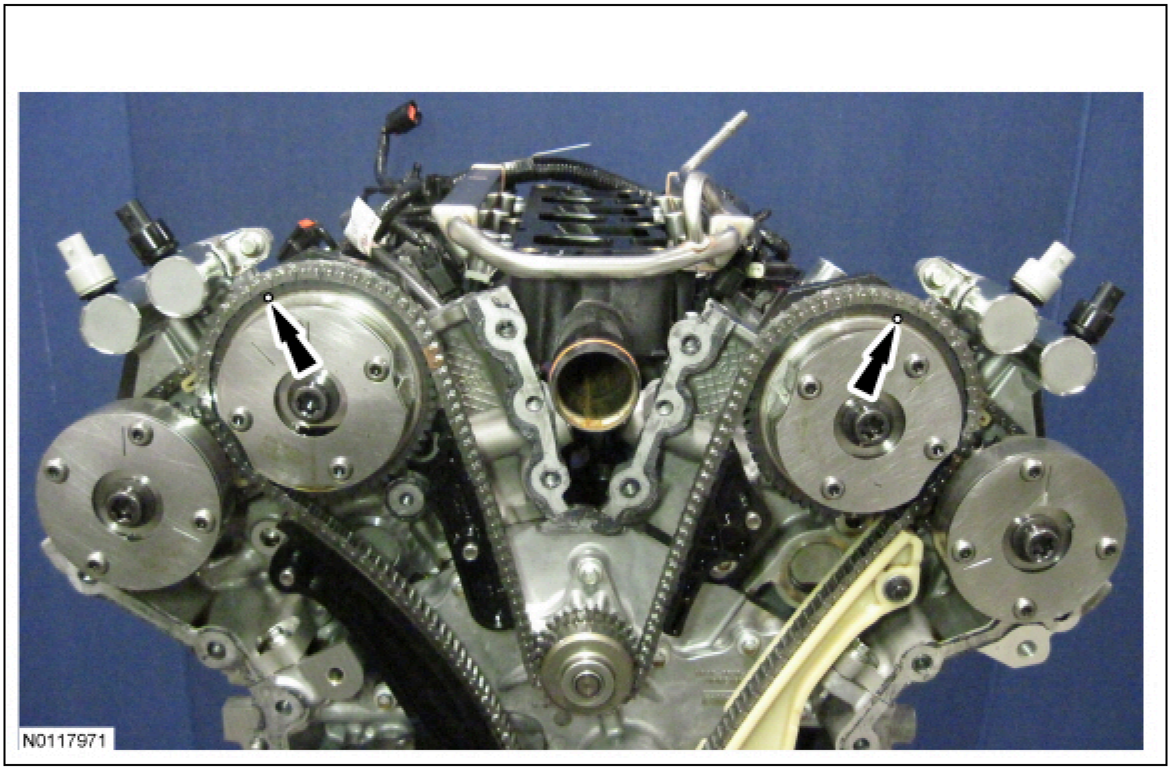
    NOTE: *Use Spindle Bearing Installer 205-150 to remove the seal.*

    If necessary, using the Spindle Bearing Installer and Handle, remove the front cover radial seal from the rear side of the front cover.

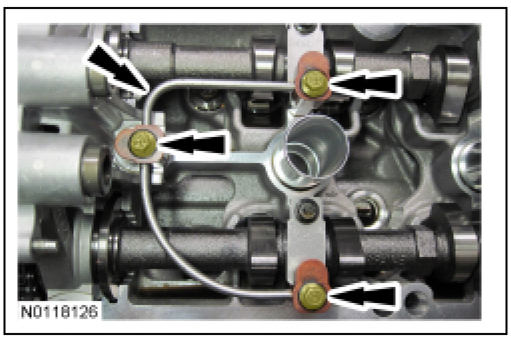


N0120096

67. Rotate the crankshaft clockwise and align the timing marks on the VCT assemblies as shown.

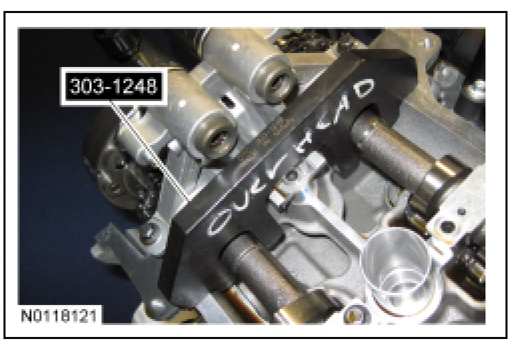


68. Remove the 3 bolts and the LH valve train oil tube.

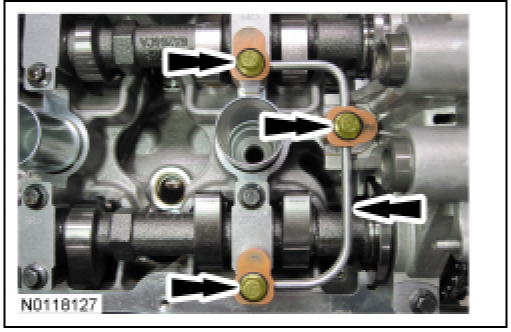


69. **NOTE:** *The Camshaft Holding Tool will hold the camshafts in the Top Dead Center (TDC) position*.

Install the Camshaft Holding Tool onto the flats of the LH camshafts.

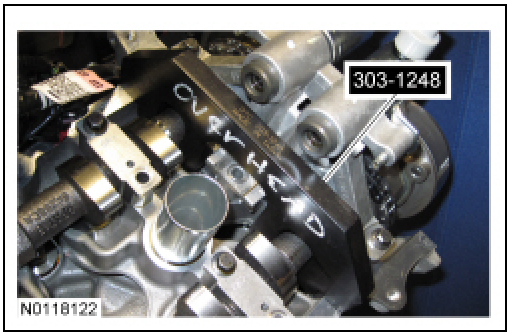


70. Remove the 3 bolts and the RH valve train oil tube.



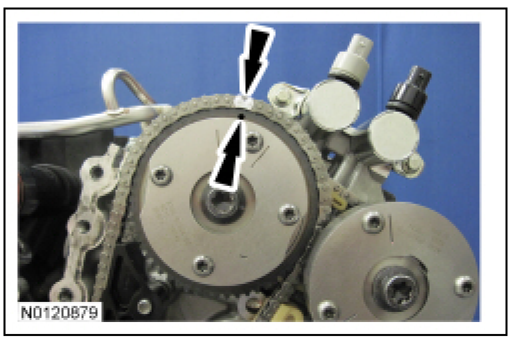
71. **NOTE:** *The Camshaft Holding Tool will hold the camshafts in the TDC position.*

    Install the Camshaft Holding Tool onto the flats of the RH camshafts.

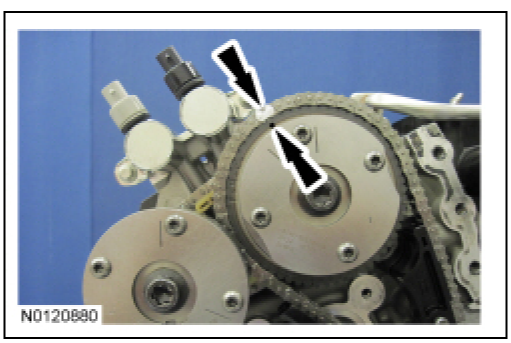


NOTE: *The following 3 steps are for primary timing chains that the colored links are not visible.*

72. Mark the timing chain link that aligns with the timing mark on the LH intake VCT assembly as shown.

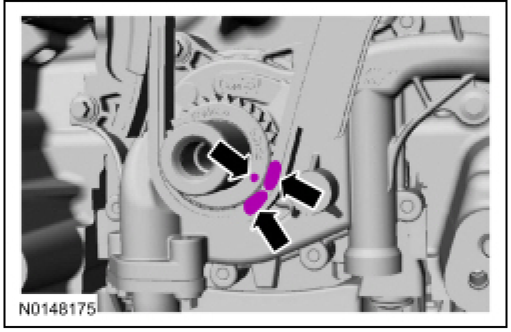


73. Mark the timing chain link that aligns with the timing mark on the RH intake VCT assembly as shown.

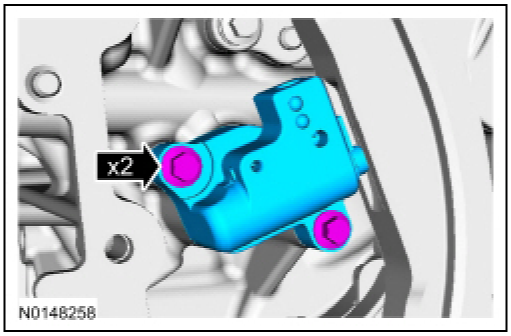


74. **NOTE:** *The crankshaft sprocket timing mark should be between the 2 colored links.*

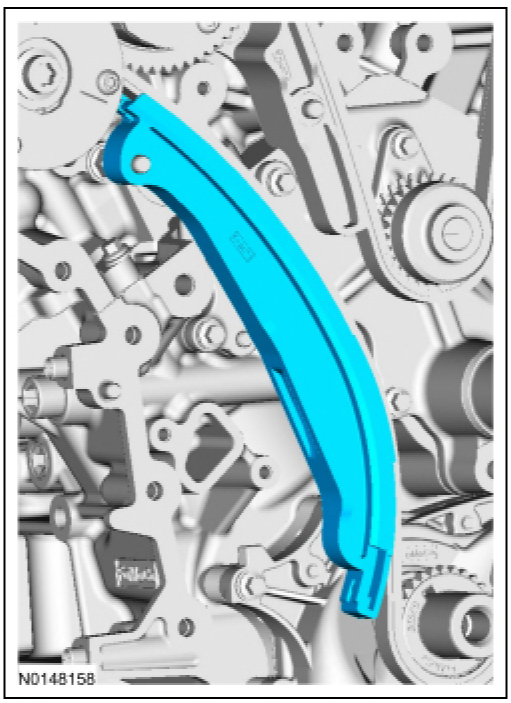
    Mark the 2 timing chain links that align with the timing mark on the crankshaft sprocket as shown.



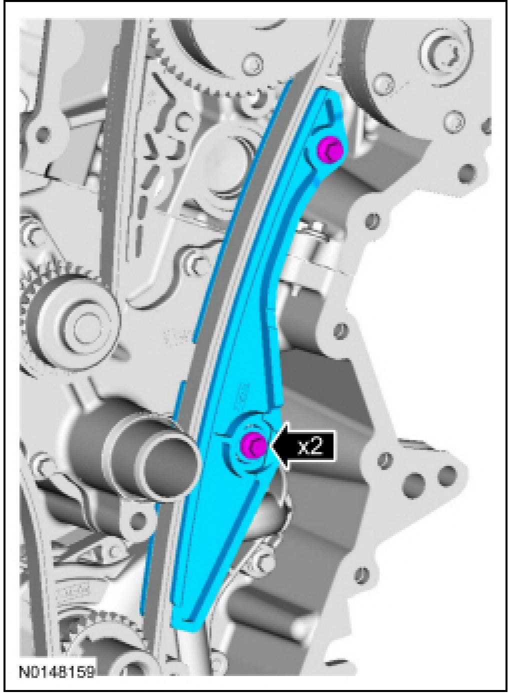
75. Remove the 2 bolts and the primary timing chain tensioner.



76. Remove the primary timing chain tensioner arm.



77. Remove the 2 bolts and the lower LH primary timing chain guide.

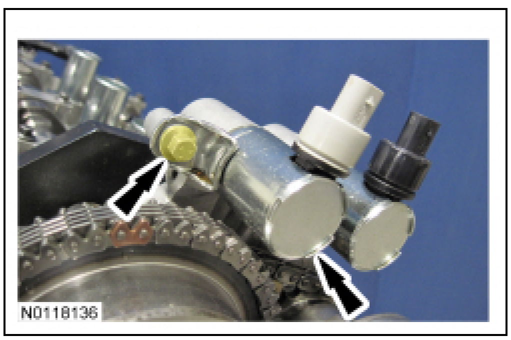


78. **NOTE:** *Removal of the VCT oil control solenoid will aid in the removal of the primary timing chain.*

    NOTE: *A slight twisting motion will aid in the removal of the VCT oil control solenoid.*

    NOTE: *Keep the VCT oil control solenoid clean of dirt and debris.*

    Remove the bolt and the LH intake VCT oil control solenoid.

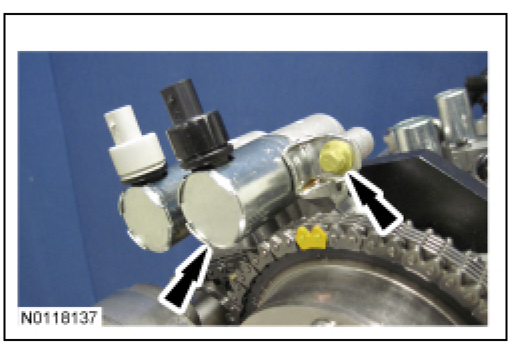


79. **NOTE:** *Removal of the VCT oil control solenoid will aid in the removal of the primary timing chain.*

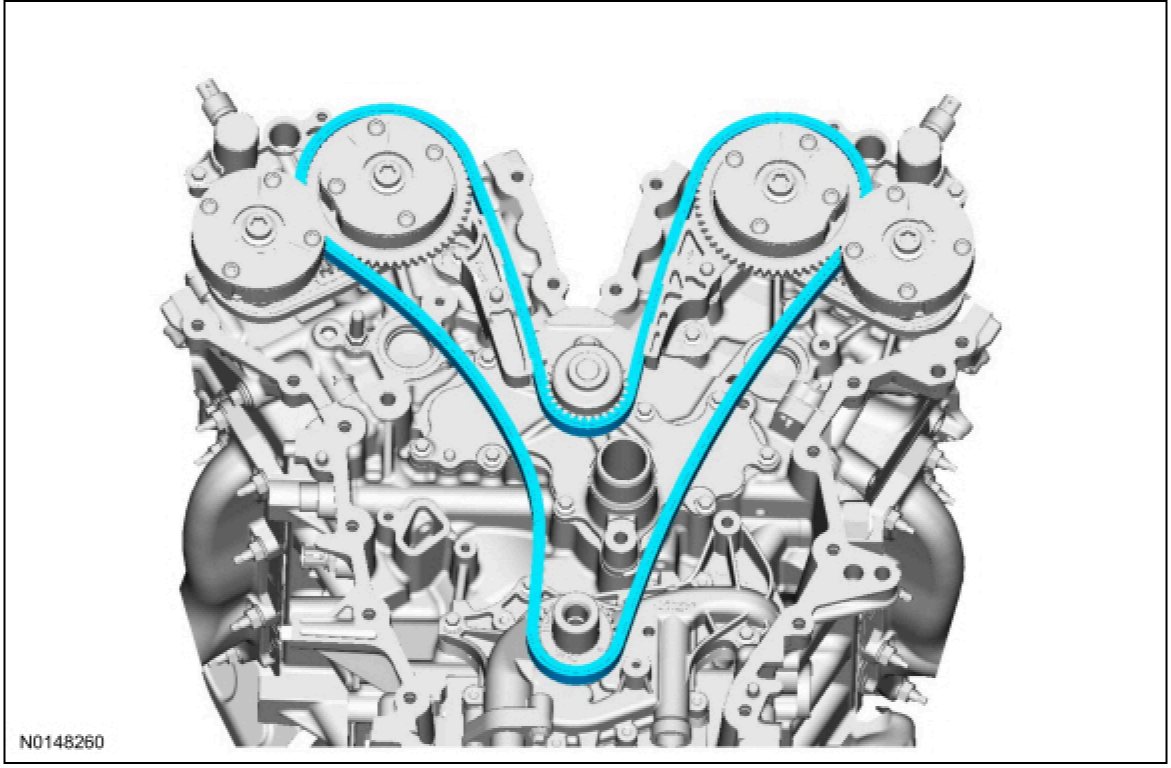
    NOTE: *A slight twisting motion will aid in the removal of the VCT oil control solenoid.*

    NOTE: *Keep the VCT oil control solenoid clean of dirt and debris.*

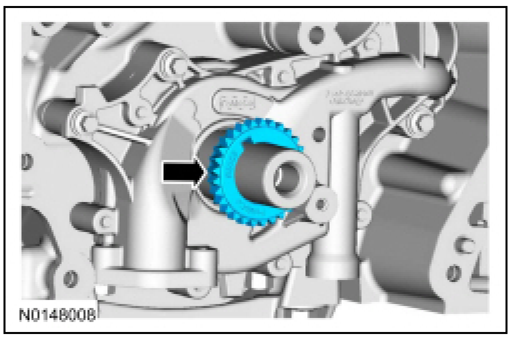
    Remove the bolt and the RH intake VCT oil control solenoid.



80. Remove the primary timing chain.

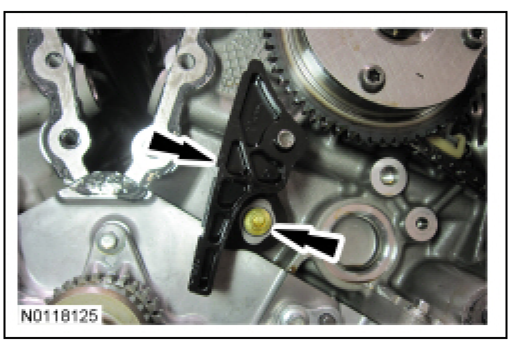


81. Remove the crankshaft timing chain sprocket.



82. **NOTICE: Do not use power tools to remove the bolt or damage to the LH primary timing chain guide may occur.**

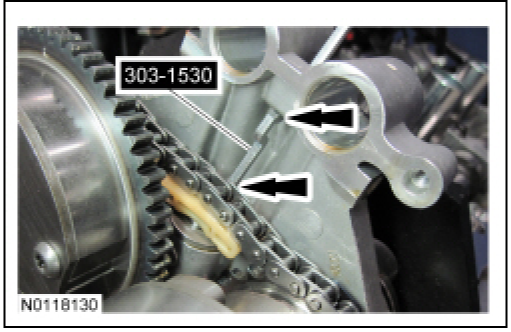
    Remove the bolt and the upper LH primary timing chain guide.



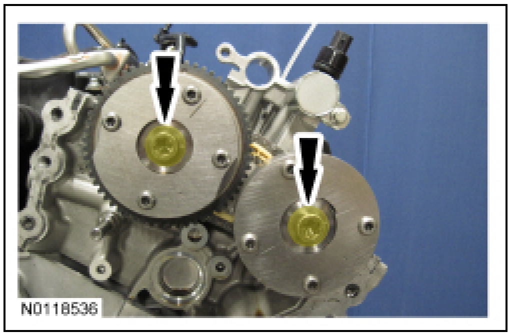
83. **NOTE:** *The 2 VCT oil control solenoids are removed for clarity.*

    NOTE: *The Secondary Chain Hold Down is inserted through a hole in the top of the mega cap.*

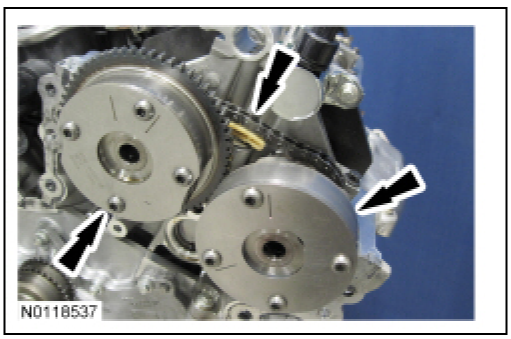
    Compress the LH secondary timing chain tensioner and install the Secondary Chain Hold Down in the hole on the rear of the secondary timing chain tensioner guide and let it hold against the mega cap to retain the tensioner in the collapsed position.



84. Remove and discard the 2 LH VCT assembly bolts.



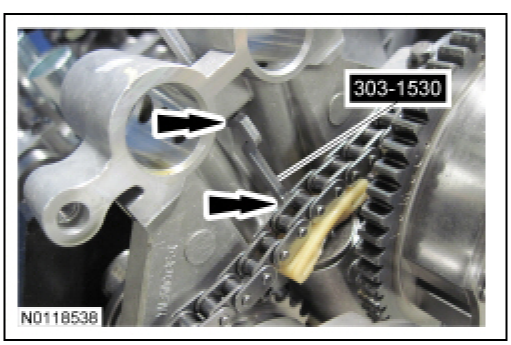
85. Remove the 2 LH VCT assemblies and secondary timing chain.



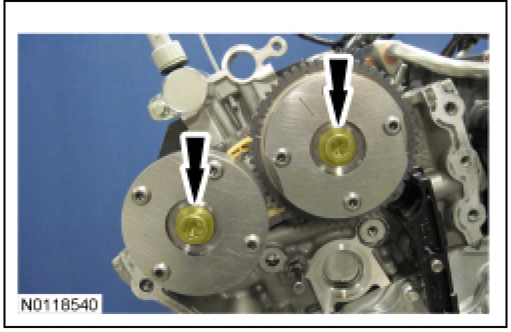
86. **NOTE:** *The 2 VCT oil control solenoids are removed for clarity.*

    NOTE: *The Secondary Chain Hold Down is inserted through a hole in the top of the mega cap.*

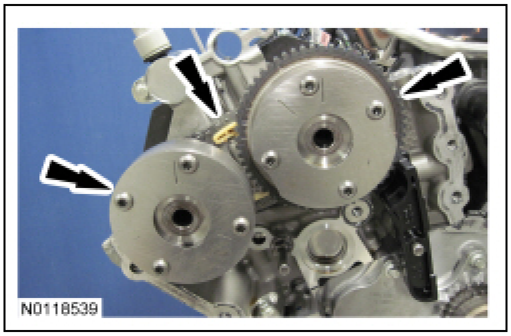
    Compress the RH secondary timing chain tensioner and install the Secondary Chain Hold Down in the hole on the rear of the secondary timing chain tensioner guide and let it hold against the mega cap to retain the tensioner in the collapsed position.



87. Remove and discard the 2 RH VCT assembly bolts.

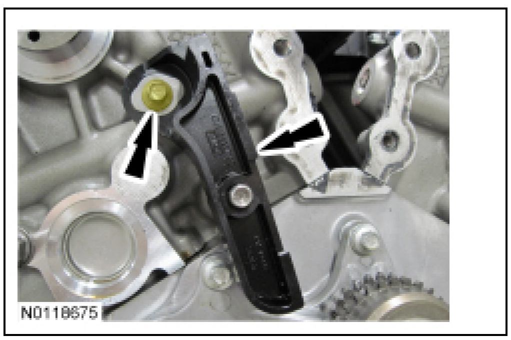


88. Remove the 2 RH VCT assemblies and secondary timing chain.

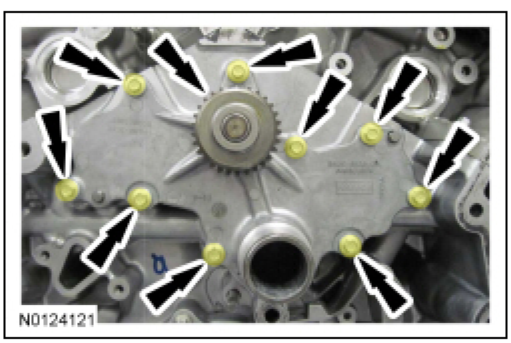


89. **NOTICE: Do not use power tools to remove the bolt or damage to the RH primary timing chain guide may occur.**

    Remove the bolt and the RH primary timing chain guide.



90. Remove the 9 bolts and the channel cover plate.
    - Discard the gaskets.



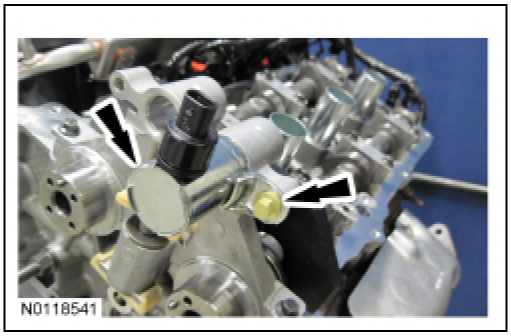
Engines with secondary timing chain tensioner removed

NOTICE: The following steps are only for the replacement of the secondary timing chain tensioners. Do not reuse the secondary timing chain tensioners if removed, or damage to the engine may occur.

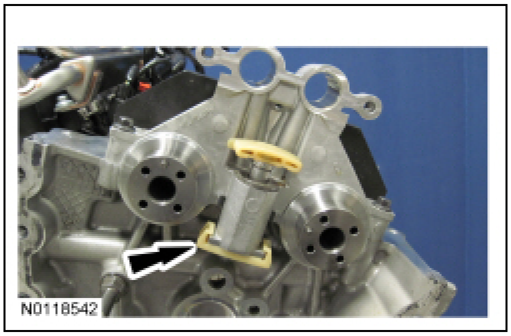
91. **NOTE:** *A slight twisting motion will aid in the removal of the VCT oil control solenoid.*

    NOTE: *Keep the VCT oil control solenoid clean of dirt and debris.*

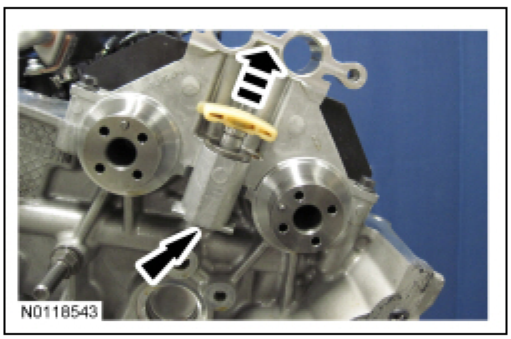
Remove the LH exhaust VCT oil control solenoid.



92. Remove the LH secondary timing chain tensioner shoe.



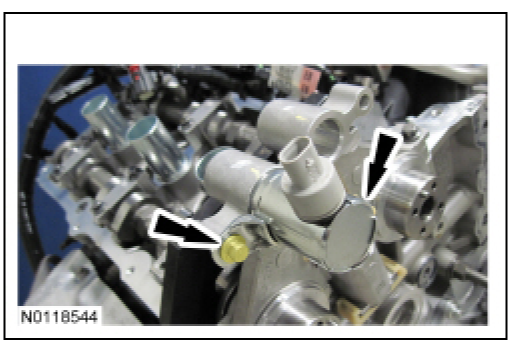
93. Remove the LH secondary timing chain tensioner by pushing up from the bottom and remove.



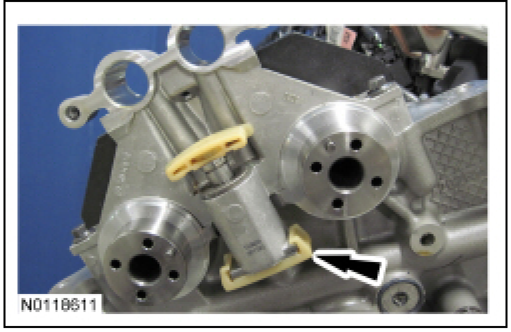
94. **NOTE:** *A slight twisting motion will aid in the removal of the VCT oil control solenoid.*

NOTE: *Keep the VCT oil control solenoid clean of dirt and debris.*

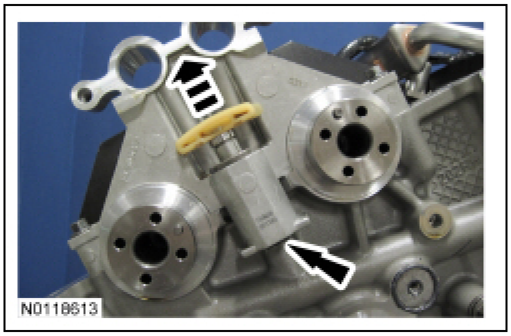
Remove the RH exhaust VCT oil control solenoid.



95. Remove the RH secondary timing chain tensioner shoe.



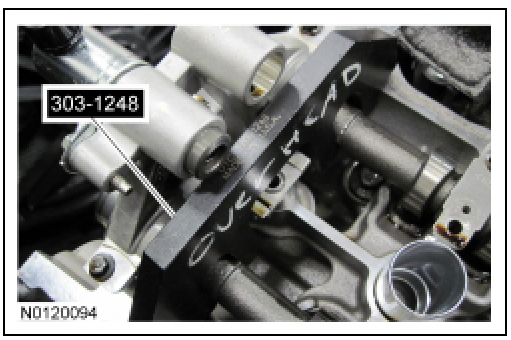
96. Remove the RH secondary timing chain tensioner by pushing up from the bottom and remove.



All engines

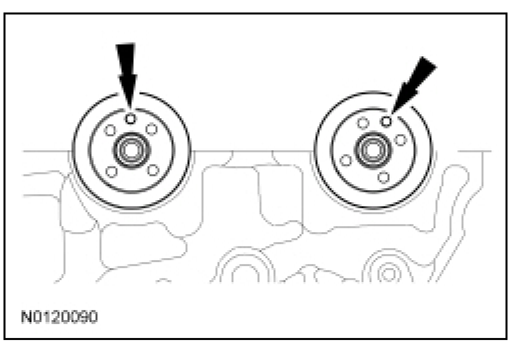
97. **NOTE:** *When the Camshaft Holding Tool is removed, valve spring pressure may rotate the LH camshafts approximately 3 degrees to a neutral position.*

    Remove the Camshaft Holding Tool from the LH camshafts.



98. **NOTICE: The camshafts must remain in the neutral position during removal or engine damage may occur.**

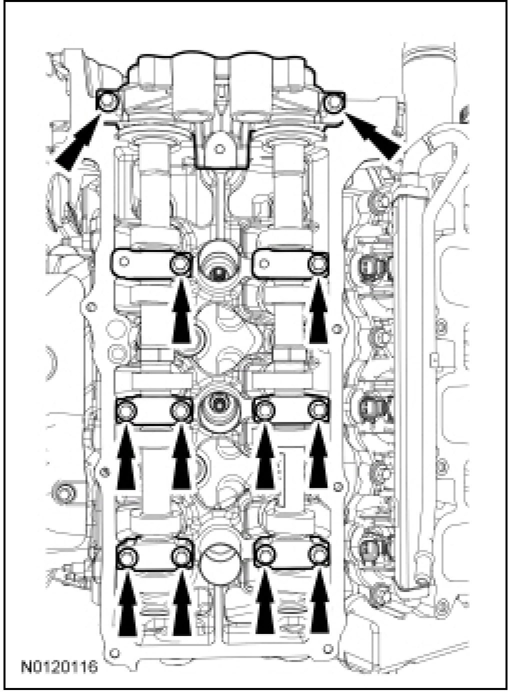
    Verify the LH camshafts are in the neutral position.



99. **NOTE:** *Cylinder head camshaft bearing caps are numbered to verify that they are assembled in their original positions.*

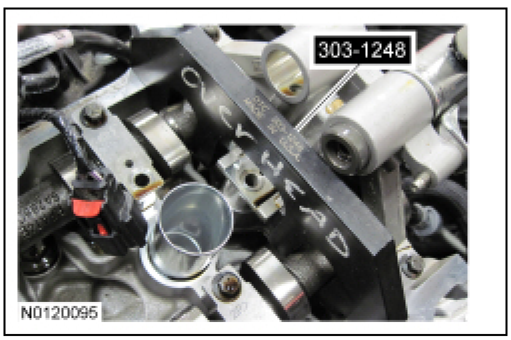
    NOTE: *Mark the exhaust and intake camshafts for installation into their original locations.*

    Remove the 12 bolts, 6 camshaft caps, mega cap and the LH camshafts.



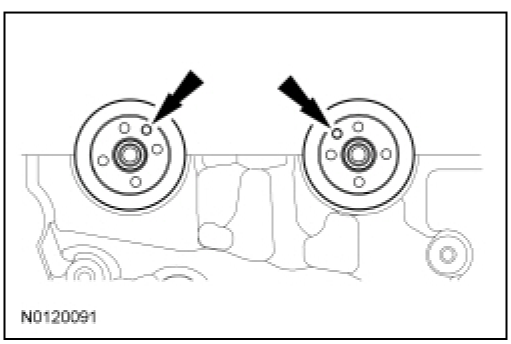
100. **NOTE:** *When the Camshaft Holding Tool is removed, valve spring pressure may rotate the RH camshafts approximately 3 degrees to a neutral position.*

    Remove the Camshaft Holding Tool from the RH camshafts.



101. **NOTICE: The camshafts must remain in the neutral position during removal or engine damage may occur.**

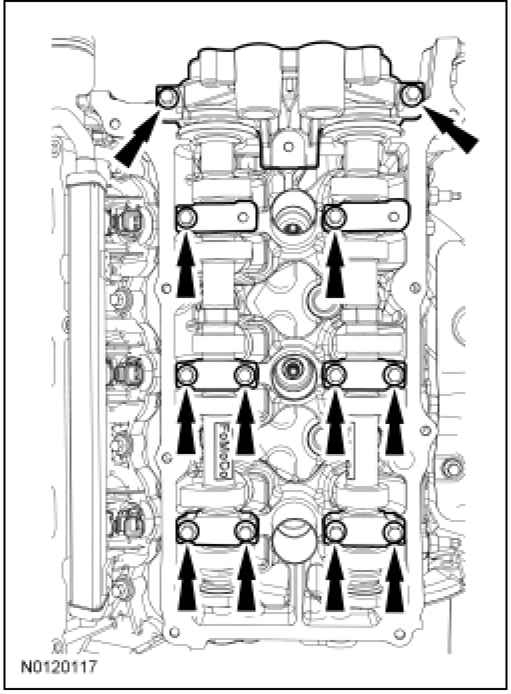
    Rotate the RH camshafts counterclockwise to the neutral position.



102. **NOTE:** *Cylinder head camshaft bearing caps are numbered to verify that they are assembled in their original positions.*

    NOTE: *Mark the exhaust and intake camshafts for installation into their original locations.*

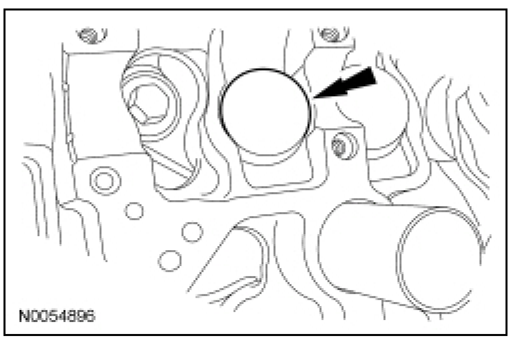
    Remove the 12 bolts, 6 camshaft caps, mega cap and the RH camshafts.



103. **NOTE:** *If the components are to be reinstalled, they must be installed in the same positions. Mark the components for installation into their original locations.*

    NOTE: *LH shown, RH similar.*

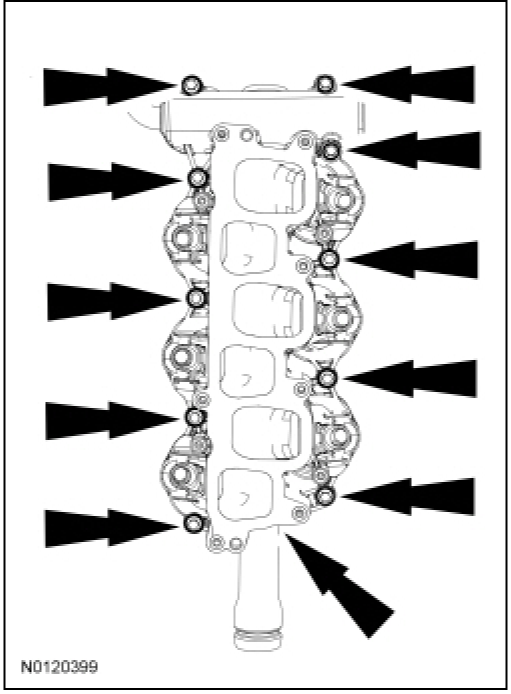
    Remove the valve tappets from the cylinder heads.



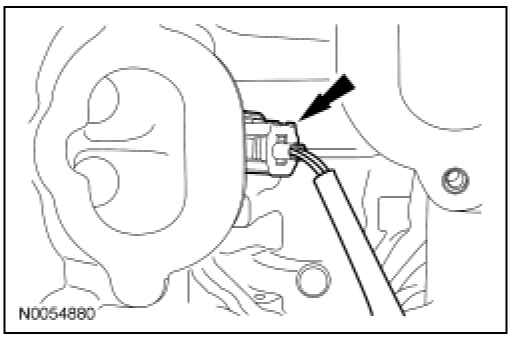
104. Inspect the valve tappets. For additional information, refer to Section 303-00.

105. **NOTICE: If the engine is repaired or replaced because of upper engine failure, typically including valve or piston damage, check the intake manifold for metal debris. If metal debris is found, install a new intake manifold. Failure to follow these instructions can result in engine damage.**

    Remove the 10 bolts and the lower intake manifold.
    - Discard the gaskets.

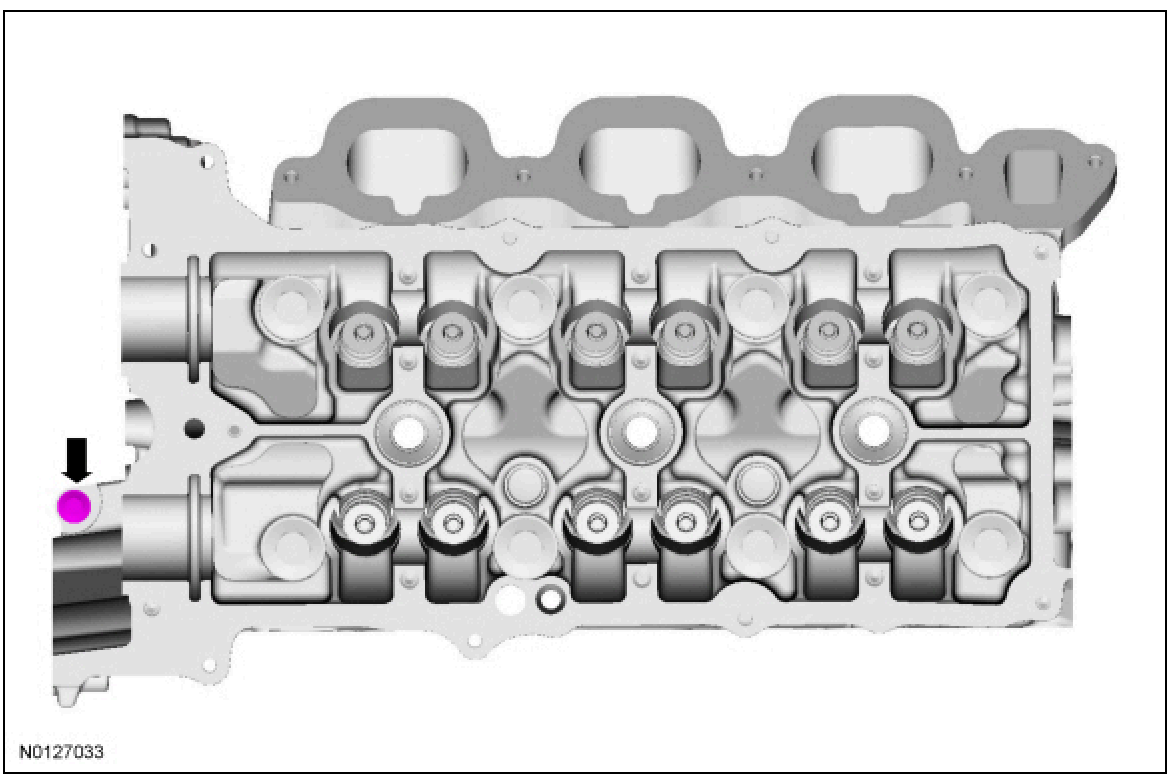


106. Disconnect and remove the CHT sensor jumper harness.



107. **NOTE:** *LH shown, RH similar.*

Remove and discard the M6 bolt from each cylinder head.



108. **NOTICE: Place clean, lint-free shop towels over exposed engine cavities. Carefully remove the towels so foreign material is not dropped into the engine. Any foreign material (including any material created while cleaning gasket surfaces) that enters the oil passages or the oil pan, may cause engine failure.**

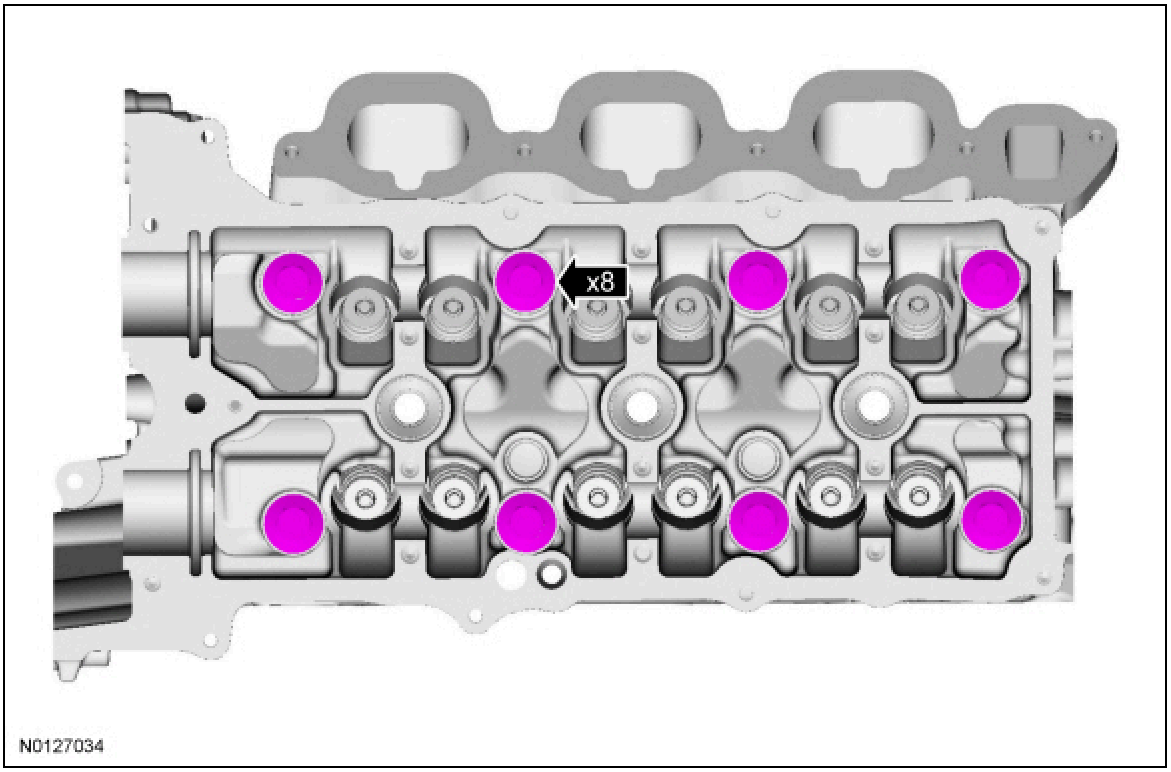
NOTICE: Aluminum surfaces are soft and may be scratched easily. Never place the cylinder head gasket surface, unprotected, on a bench surface.

NOTE: *The cylinder head bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.*

NOTE: *LH shown, RH similar.*

Remove and discard the 8 bolts from each cylinder head.

- Remove the cylinder heads.
- Discard the cylinder head gaskets.



109. **NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges that make leak paths. Use a plastic scraping tool to remove all traces of the head gasket.**

NOTE: *Observe all warnings or cautions and follow all application directions contained on the packaging of the silicone gasket remover and the metal surface prep.*

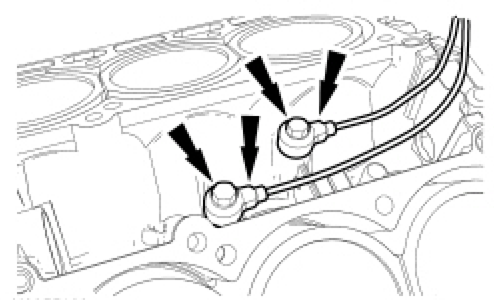
NOTE: *If there is no residual gasket material present, metal surface prep can be used to clean and prepare the surfaces.*

Clean the cylinder head-to-cylinder block mating surfaces of both the cylinder heads and the cylinder block in the following sequence.

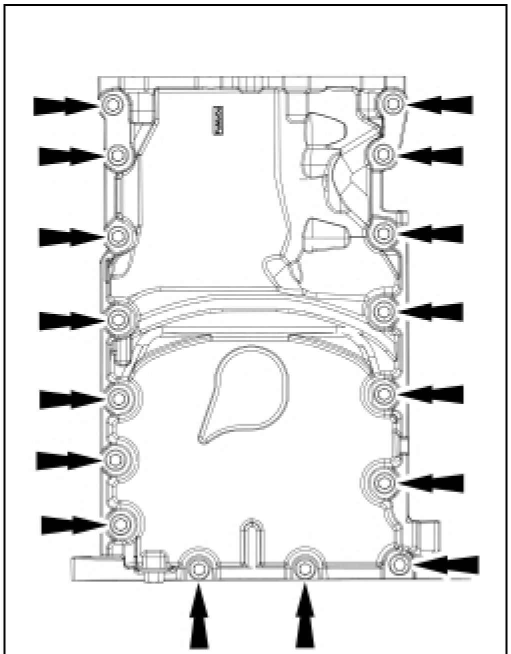
1. Remove any large deposits of silicone or gasket material with a plastic scraper.
2. Apply silicone gasket remover, following package directions, and allow to set for several minutes.
3. Remove the silicone gasket remover with a plastic scraper. A second application of silicone gasket remover may be required if residual traces of silicone or gasket material remain.
4. Apply metal surface prep, following package directions, to remove any remaining traces of oil or coolant and to prepare the surfaces to bond with the new gasket. Do not attempt to make the metal shiny. Some staining of the metal surfaces is normal.

110. Support the cylinder head on a bench with the head gasket side up. Check the cylinder head distortion and the cylinder block distortion. For additional information, refer to Section 303-00.

111. Remove the 2 bolts and the KS .

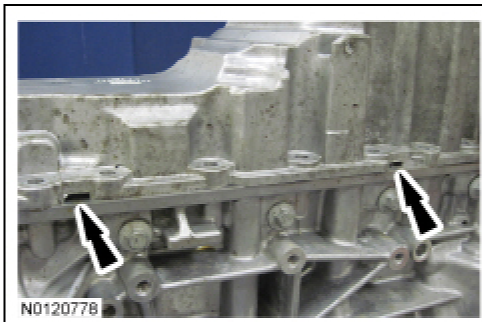

N0055160

112. Remove the 16 oil pan bolts.

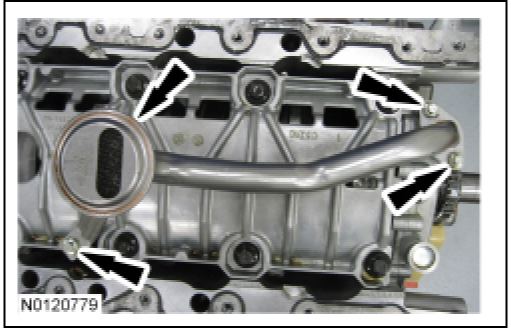

N0114753

113. **NOTE:** *RH shown, LH similar.*

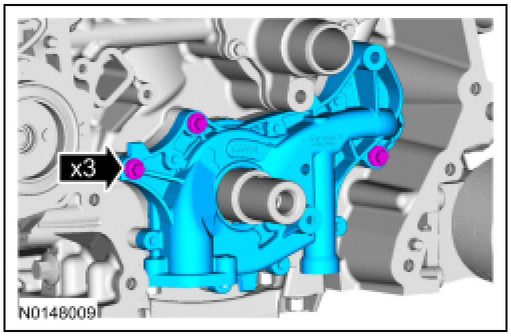
Using a pry tool, locate the 2 pry pads at the LH and RH side of the oil pan and pry the oil pan loose and remove.


N0120778

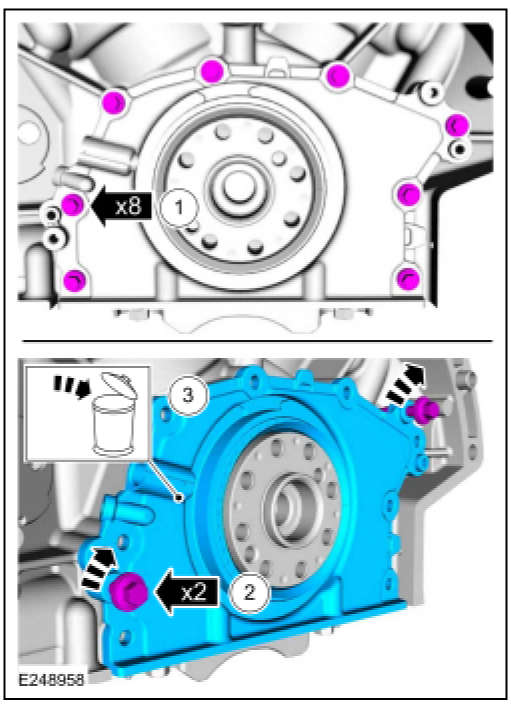
114. Remove the 3 bolts and the oil pump screen and the pickup tube.
- Discard the O-ring seal.



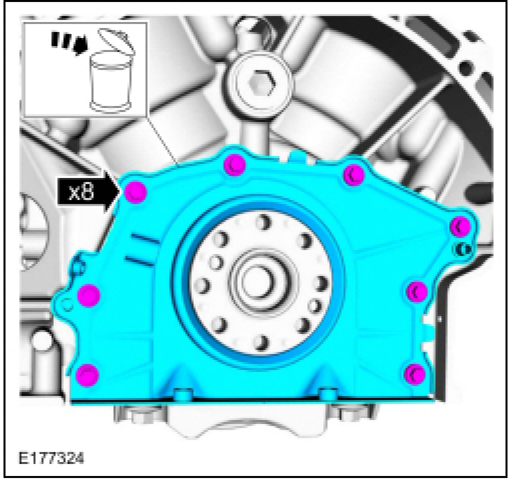
115. Remove the 3 bolts and the oil pump.



116. If equipped with a cast aluminum rear seal retainer plate (early build), remove the crankshaft rear seal retainer plate as follows:
    1. Remove the bolts.
    2. Install the 2 M6 oil pan bolts (finger-tight) into the 2 threaded holes in the crankshaft rear seal retainer plate.
        - Alternately tighten the 2 bolts one turn at a time until the crankshaft rear seal retainer plate-to-cylinder block seal is released.
        - Remove the 2 M6 oil pan bolts.
    3. Remove and discard the crankshaft rear seal retainer plate.



117. If equipped with a stamped steel rear seal retainer plate (late build), remove the bolts and the crankshaft rear seal and retainer plate.
    - Discard the rear seal and retainer plate.



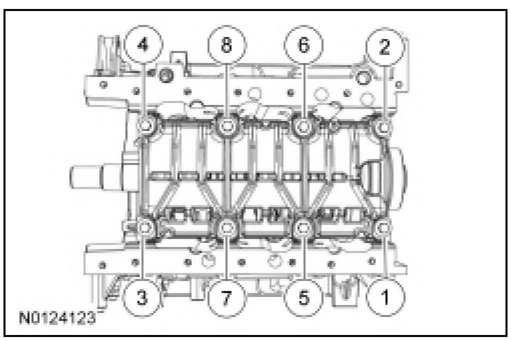
118. **NOTICE: Only use a 3M™ Roloc® Bristle Disk (2-in white, part number 07528) to clean the engine front cover and oil pan. Do not use metal scrapers, wire brushes or any other power abrasive disk to clean the engine front cover and oil pan. These tools cause scratches and gouges that make leak paths.**

    Clean the engine front cover and oil pan using a 3M Roloc® Bristle Disk (2 inch, white, part number 07528) in a suitable tool turning at the recommended speed of 15,000 rpm. Thoroughly wash the engine front cover and oil pan to remove any foreign material, including any abrasive particles created during the cleaning process.

119. Before removing the pistons, inspect the top of the cylinder bores. If necessary, remove the ridge or carbon deposits from each cylinder using an abrasive pad or equivalent, following manufacturer's instructions.

120. **NOTE:** *The main bearing cap support brace bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.*

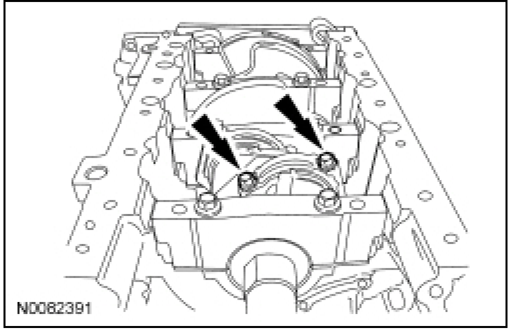
    Remove the bolts in the sequence shown.
    - Remove the main bearing cap support brace.
    - Discard the bolts.



121. **NOTE:** *The connecting rod cap bolts are a tighten-to-yield design. The original connecting rod cap bolts will be used when measuring the connecting rod large end bore during assembly. The connecting rod cap bolts will be discarded after measurement.*

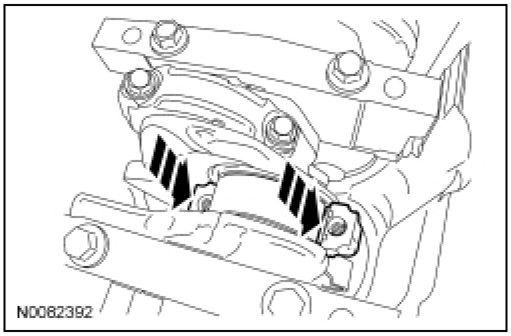
    NOTE: *Clearly mark the position and orientation of the connecting rods, connecting rod caps and connecting rod bearings for reassembly.*

    Remove the connecting rod cap bolts and cap.



122. **NOTICE: Do not scratch the cylinder walls or crankshaft journals with the connecting rod.**

 Remove the piston/rod assembly from the engine block.

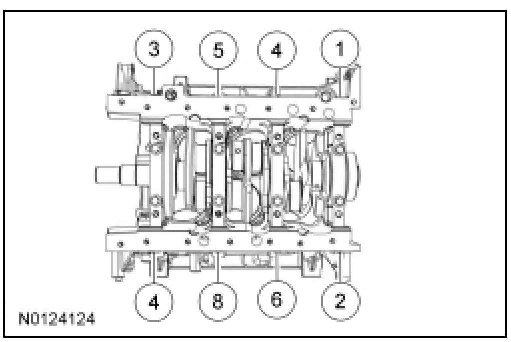


123. Repeat the previous 2 steps until all the piston/rod assemblies are removed from the engine block.

124. **NOTE:** *The 8 main bearing cap side bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.*

 NOTE: *Clearly mark the position and orientation of the main bearing caps for reassembly.*

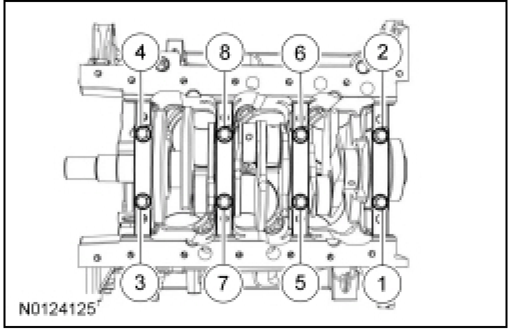
 Remove the 8 main bearing cap side bolts in the sequence shown.
 - Discard the bolts.



125. **NOTE:** *The 8 main bearing cap bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.*

 NOTE: *Clearly mark the position and orientation of the main bearing caps for reassembly.*

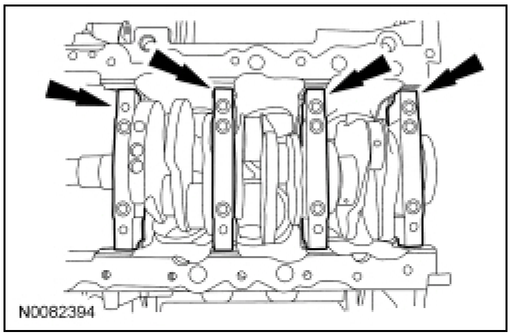
 Remove the 8 main bearing cap bolts in the sequence shown.
 - Discard the bolts.



126. **NOTE:** *If the main bearings are being reused, mark them for correct position and orientation for reassembly.*

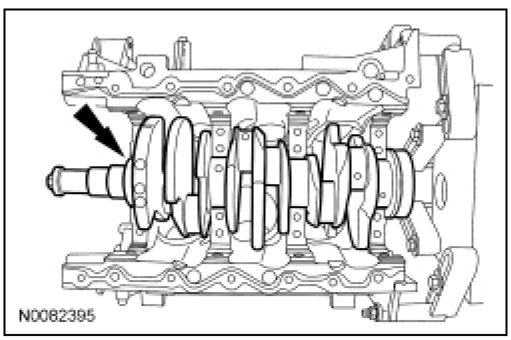
NOTE: *Note the position of the thrust washer on the outside of the No. 4 rear main bearing cap.*

Remove the 4 main bearing caps.



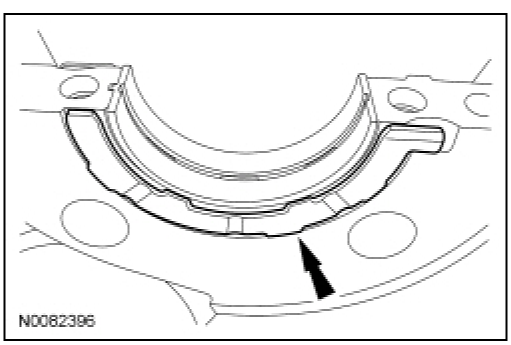
127. **NOTE:** *Note the position of the 2 thrust washers on the inside and outside of the rear main bearing bulkhead.*

Remove the crankshaft.



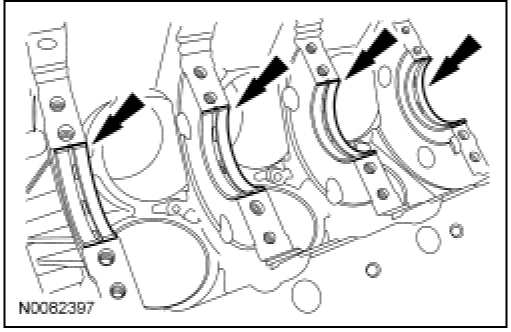
128. **NOTE:** *Inside shown, outside similar.*

Remove the 2 crankshaft thrust bearings from the rear main bearing bulkhead.



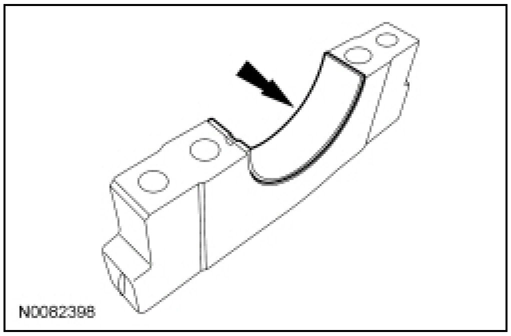
129. **NOTE:** *If the main bearings are being reused, mark them for correct position and orientation for reassembly.*

Remove the 4 crankshaft main bearings from the cylinder block.



130. **NOTE:** *If the main bearings are being reused, mark them for correct position and orientation for reassembly.*

Remove the 4 crankshaft main bearings from the main bearing caps.



131. Inspect the cylinder block, bearing cap support brace, pistons and connecting rods. For additional information, refer to Section 303-00.

132. **NOTICE: Do not use wire brushes, power abrasive discs or 3M™ Roloc® Bristle Disk (2-in white, part number 07528) to clean the sealing surfaces. These tools cause scratches and gouges that make leak paths. They also cause contamination that causes premature engine failure. Remove all traces of the gasket.**

Clean the sealing surfaces of the cylinder heads, the cylinder block and the oil pan in the following sequence.

1. Remove any large deposits of silicone or gasket material.
2. Apply silicone gasket remover and allow to set for several minutes.
3. Remove the silicone gasket remover. A second application of silicone gasket remover may be required if residual traces of silicone or gasket material remain.
4. Apply metal surface prep to remove any remaining traces of oil and to prepare the surfaces to bond. Do not attempt to make the metal shiny. Some staining of the metal surfaces is normal.
5. Make sure the 2 locating dowel pins are seated correctly in the cylinder block.